Chapter 5 – Seakeeping Models for Ships and USVs

The study of ship dynamics is traditionally divided into two main theories: maneuvering and seakeeping.

Seakeeping refers to the motion of marine craft on a constant course and speed in waves (including the trivial case of zero speed). In seakeeping analysis, a dissipative force representing fluid memory effects (Cummins, 1962) is introduced due to the frequency dependence of the potential coefficients (added mass and damping).

Maneuvering refers to the study of ship motion in the absence of wave excitation (calm water). The equations of motion (**Chapter 6**) are formulated using equivalent constant added mass and damping matrices, defined for a given sea state to preserve kinetic energy and power.







Cummins, W. E. (1962). The Impulse Response Function and Ship Motions. Techn. Report 1661. *David Taylor Model Basin. Hydrodynamics Laboratory*, USA.

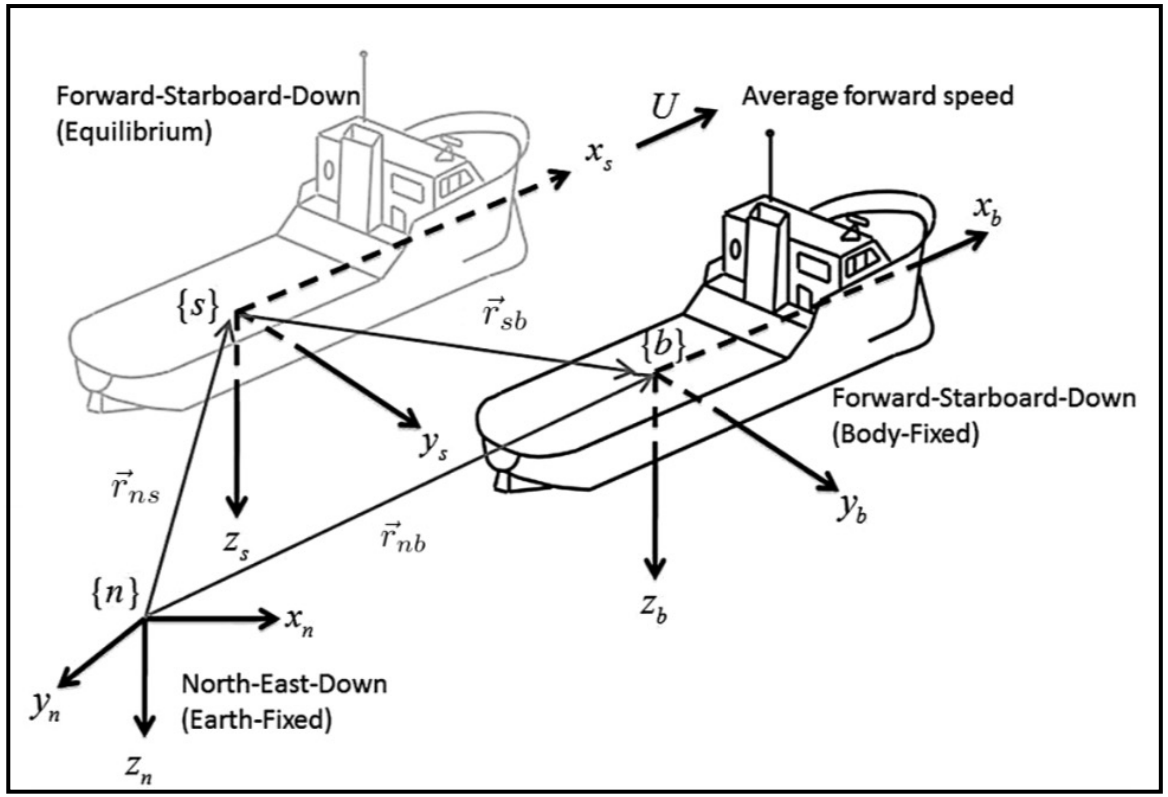
Chapter Goals

- Distinguish frequency-domain seakeeping models from time-domain maneuvering models and explain when each is appropriate.
- Define the seakeeping perturbation coordinates in {s} and transform motions and forces between {s} and the BODY-fixed frame {b}.
- Interpret frequency-dependent added-mass and potential-damping matrices, hydrostatic restoring forces, and viscous-damping contributions.
- Explain how frequency-dependent added mass and damping are identified from forced-oscillation experiments using frequency-by-frequency least-squares fitting or computed using potential-flow programs.
- Explain why inserting frequency-dependent hydrodynamic coefficients directly into a time-domain differential equation causes an inconsistent mixing of time- and frequency-domain quantities.
- Formulate Cummins' equation using asymptotic added mass and retardation functions to represent fluid-memory effects.
- Explain when zero-frequency coefficients are reasonable for surge, sway, and yaw, and why this approximation can fail for coupled 6-DOF motion.
- Compute sea-state-dependent equivalent added-mass and damping matrices by spectral power-based averaging and use them in a BODY-fixed time-domain model.

Introduction to Seakeeping Models

Seakeeping Kinematics

The SEAKEEPING reference frame {s} is not fixed to the craft; It is fixed to the equilibrium position and orientation of a ship moving at constant forward speed U and mean heading angle $\overline{\psi}$.
Without waves, {b} and {s} coincide. In waves, {b} oscillates about {s}, so the difference between them represents the wave-induced motion.



Seakeeping Perturbation Coordinates

$$\boldsymbol{\xi} = [x^n - \bar{x}^n,\ y^n - \bar{y}^n,\ z^n - \bar{z}^n,\ \phi,\ \theta,\ \psi - \bar{\psi}]^\top.$$

$$\delta\boldsymbol{\nu} := \boldsymbol{\nu} - \bar{\boldsymbol{\nu}} \qquad \delta\boldsymbol{\nu} = [\delta u,\ \delta v,\ \delta w,\ \delta p,\ \delta q,\ \delta r]^\top$$

$$\bar{\boldsymbol{\nu}} = [U,\ 0,\ 0,\ 0,\ 0,\ 0]^\top.$$

$$\boldsymbol{\xi} := \delta\boldsymbol{\eta} = \boldsymbol{\eta} - \bar{\boldsymbol{\eta}}$$

$$\dot{\boldsymbol{\xi}} = \delta\boldsymbol{\nu}, \qquad \ddot{\boldsymbol{\xi}} = \delta\dot{\boldsymbol{\nu}}$$

$$\boldsymbol{v}_{ns}^n = [U\cos(\bar{\psi}),\ U\sin(\bar{\psi}),\ 0]^\top$$
$$\boldsymbol{\omega}_{ns}^n = [0,\ 0,\ 0]^\top$$
$$\boldsymbol{\Theta}_{ns} = [0,\ 0,\ \bar{\psi}]^\top$$

Classical Frequency-Domain Seakeeping Model used in Naval Architecture

$$(\boldsymbol{M}_{RB} + \boldsymbol{A}(\omega))\,\ddot{\boldsymbol{\xi}} + (\boldsymbol{B}(\omega) + \boldsymbol{B}_v)\,\dot{\boldsymbol{\xi}} + \boldsymbol{C}\boldsymbol{\xi} = \delta\boldsymbol{\tau} + \boldsymbol{\tau}_{\text{wind}} + \boldsymbol{\tau}_{\text{wave}} + \boldsymbol{\tau}_{\text{current}}$$

Later, we will transform the frequency-domain model from {s} to {b}.

Classical Seakeeping Model used in Naval Architecture

$$(\boldsymbol{M}_{RB} + \boldsymbol{A}(\omega))\,\ddot{\boldsymbol{\xi}} + (\boldsymbol{B}(\omega) + \boldsymbol{B}_v)\,\dot{\boldsymbol{\xi}} + \boldsymbol{C}\boldsymbol{\xi} = \delta\boldsymbol{\tau} + \boldsymbol{\tau}_{\text{wind}} + \boldsymbol{\tau}_{\text{wave}} + \boldsymbol{\tau}_{\text{current}}$$

- $\boldsymbol{\xi}$ — Generalized position and orientation
- $\mathbf{M}_{RB}$ — Rigid-body mass and inertia
- $\mathbf{A}(\omega)$ — Frequency-dependent added mass
- $\mathbf{B}(\omega)$ — Frequency-dependent radiation damping
- $\mathbf{B}_v$ — Viscous damping
- $\mathbf{C}$ — Hydrostatic restoring
- $\delta\,\boldsymbol{\tau}$ — Control forces and moments
- $\boldsymbol{\tau}_{\text{wind}}$ — Wind loads
- $\boldsymbol{\tau}_{\text{wave}}$ — Wave-excitation loads
- $\boldsymbol{\tau}_{\text{current}}$ — Current-induced loads

The matrices ***A***(ω) and ***B***(ω) are frequency dependent.

We will discuss three approaches for time-domain simulation:

- **Cummins' equation**
- **Zero-frequency approximation**
- **Power-based averaging**

This model is well established in the hydrodynamics literature. However, the commonly used "time-domain" form is not a true ODE because it mixes time- and frequency-domain quantities through an abuse of notation. Hence, the model is not an ODE depending only on time, but an integro-differential equation.

Frequency-Dependent Hydrodynamic Coefficients

Frequency-Dependent Hydrodynamic Coefficients

$$\boldsymbol{M}_{RB}\ddot{\boldsymbol{\xi}} = \delta\boldsymbol{\tau} + \boldsymbol{\tau}_{\text{hyd}} + \boldsymbol{\tau}_{\text{hs}} + \boldsymbol{\tau}_{\text{wave}} \qquad \delta\boldsymbol{\tau} = \mathbf{0}$$

$$(\boldsymbol{M}_{RB} + \boldsymbol{A}(\omega))\,\ddot{\boldsymbol{\xi}} + (\boldsymbol{B}(\omega) + \boldsymbol{B}_v)\,\dot{\boldsymbol{\xi}} + \boldsymbol{C}\boldsymbol{\xi} = \boldsymbol{\tau}_{\text{wave}}$$

- $\mathbf{A}(\omega)$: added mass matrix
- $\mathbf{B}(\omega)$: potential damping matrix
- $\mathbf{B}_v$: constant viscous damping matrix
- $\mathbf{C}$: spring stiffness or hydrostatic matrix

The hydrodynamic and hydrostatic forces are:

$$\boldsymbol{\tau}_{\text{hyd}} = -\underbrace{\boldsymbol{A}(\omega)\ddot{\boldsymbol{\xi}}}_{\substack{\text{added-mass}\\ \text{force}}} - \underbrace{\boldsymbol{B}(\omega)\dot{\boldsymbol{\xi}}}_{\substack{\text{potential}\\ \text{damping}}} - \underbrace{\boldsymbol{B}_v\dot{\boldsymbol{\xi}}}_{\substack{\text{viscous}\\ \text{damping}}}$$

$$\boldsymbol{\tau}_{\text{hs}} = -\underbrace{\boldsymbol{C}\boldsymbol{\xi}}_{\substack{\text{restoring}\\ \text{force}}}$$

1st- and 2nd-order wave-induced excitation forces are added by linear superposition (**Chapter 10**) using Response Amplitude Operators (RAOs)

The model represent a "**Hydrodynamic Mass–Damper–Spring System**" which varies with the frequency ω of the forced oscillation.

Definition 5.1 (Added mass) *Hydrodynamic added mass is the apparent increase in inertia caused by the need to accelerate the surrounding fluid together with the vessel. Hence, an accelerating vessel behaves as if its rigid-body mass and inertia are larger than their physical values.*

Definition 5.2 (Potential damping) *Potential damping is the hydrodynamic damping caused by energy radiated away from the vessel in the form of surface waves. It is therefore also referred to as radiation damping or wave-radiation damping.*

Added mass should not be interpreted as a fixed amount of water being carried or dragged along with the vessel. Instead, added mass is a dynamic effect arising from the pressure field generated by the accelerating motion, and its value therefore depends on the geometry of the body and the oscillation frequency.

Forced-Oscillation Experiment

$$\boldsymbol{M}_{RB}\ddot{\boldsymbol{\xi}} = \delta\boldsymbol{\tau} + \boldsymbol{\tau}_{\text{hyd}} + \boldsymbol{\tau}_{\text{hs}} + \boldsymbol{\tau}_{\text{wave}}$$

Wave excitation

$$\boldsymbol{\tau}_{\text{wave}} = \boldsymbol{f}_{\text{wave}} \cos(\omega t)$$

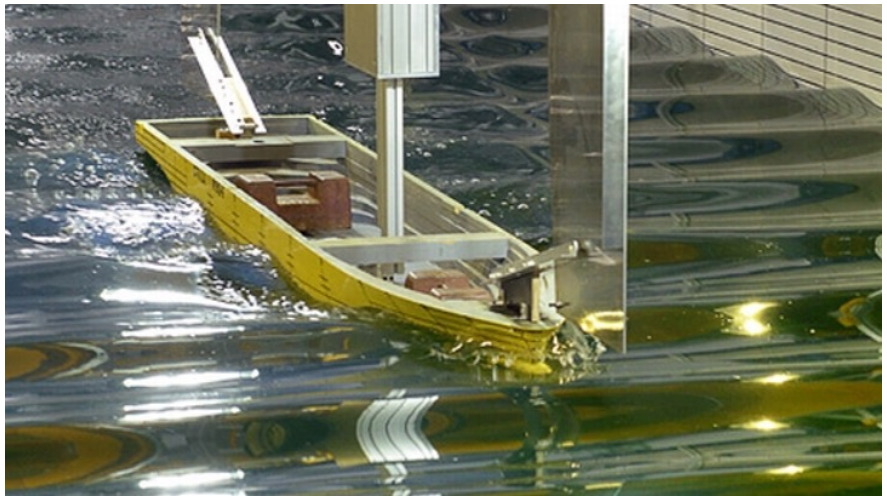
In an experimental setup with a restrained scale model, it is possible to vary the wave excitation frequency ω and the amplitudes $f_{\text{wave, i}}$ for $i = 1...6$ of the excitation force. Hence, by measuring the position and attitude vector ξ_i the response of the second-order system can be fitted to a linear model (e.g. by least squares) for each frequency ω.

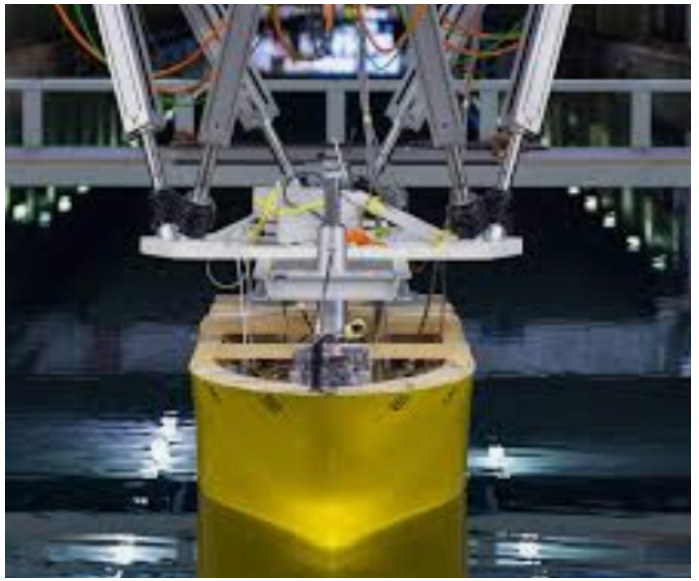
$$\left[\boldsymbol{M}_{RB} + \boldsymbol{A}(\omega)\right]\ddot{\boldsymbol{\xi}} + \boldsymbol{B}_{\text{total}}(\omega)\dot{\boldsymbol{\xi}} + \boldsymbol{C}\boldsymbol{\xi} = \boldsymbol{\tau}_{wave}$$

$$\boldsymbol{B}_{\text{total}}(\omega) = \boldsymbol{B}(\omega) + \boldsymbol{B}_v$$

Note that the resulting differential is mixing time and frequency since the model has frequency-dependent matrices

This gives two frequency-dependent model matrices $\boldsymbol{A}(\omega)$ and $\boldsymbol{B}_{\text{total}}(\omega)$.





Potential Theory Programs and CFD

Potential-theory programs model water flow around ships and offshore structures, assuming the fluid is inviscid, incompressible, and irrotational. Alternatives are viscous CFD (RANS), which includes friction, turbulence, and flow separation.

They programs computes wave patterns, hydrodynamic loads, added mass, damping, and vessel motions.

$$\left[\boldsymbol{M}_{RB} + \boldsymbol{A}(\omega)\right]\ddot{\boldsymbol{\xi}} + \boldsymbol{B}_{\text{total}}(\omega)\dot{\boldsymbol{\xi}} + \boldsymbol{C}\boldsymbol{\xi} = \boldsymbol{\tau}_{wave}$$

$$\boldsymbol{B}_{\text{total}}(\omega) = \boldsymbol{B}(\omega) + \boldsymbol{B}_v$$

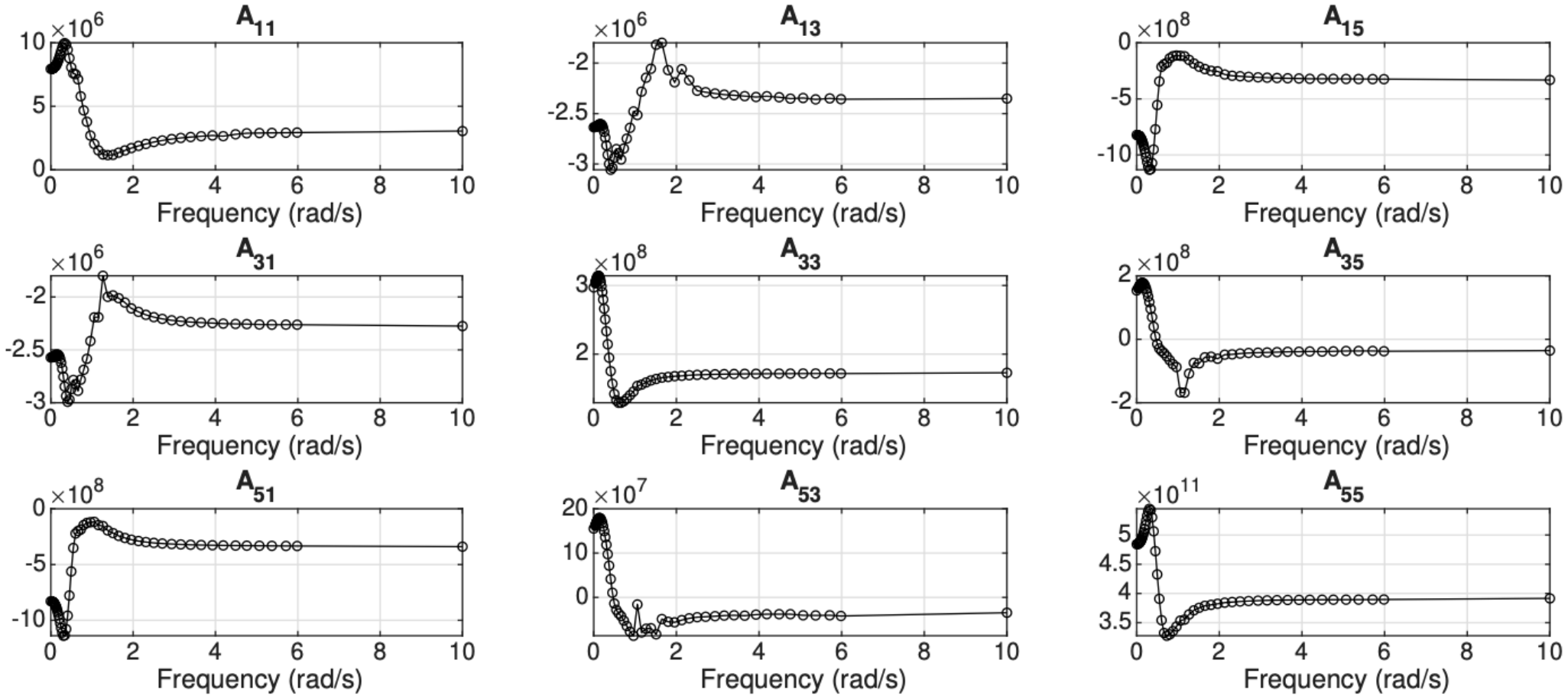
You should add viscous damping manually, if not computed.

Potential theory programs typically compute:

- Frequency-dependent added mass matrix, $\boldsymbol{A}(\omega)$
- Frequency-dependent potential damping matrix, $\boldsymbol{B}(\omega)$
- Constant restoring matrix, $\boldsymbol{C}$
- 1st- and 2nd-order wave-induced forces and motions (amplitudes and phases) for given wave directions and frequencies
- … and much more

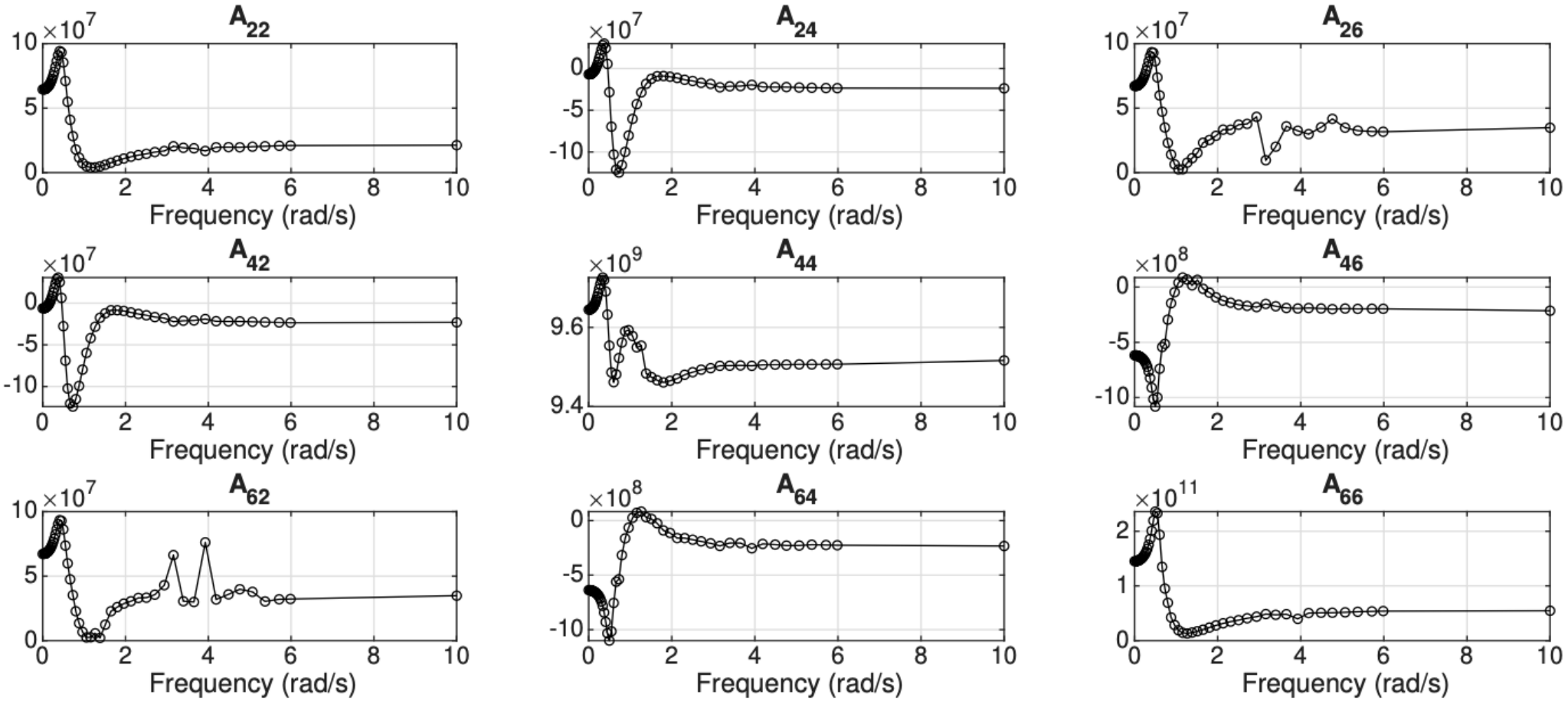
However, potential theory codes give you no viscous damping terms

Longitudinal Added-Mass Coefficients



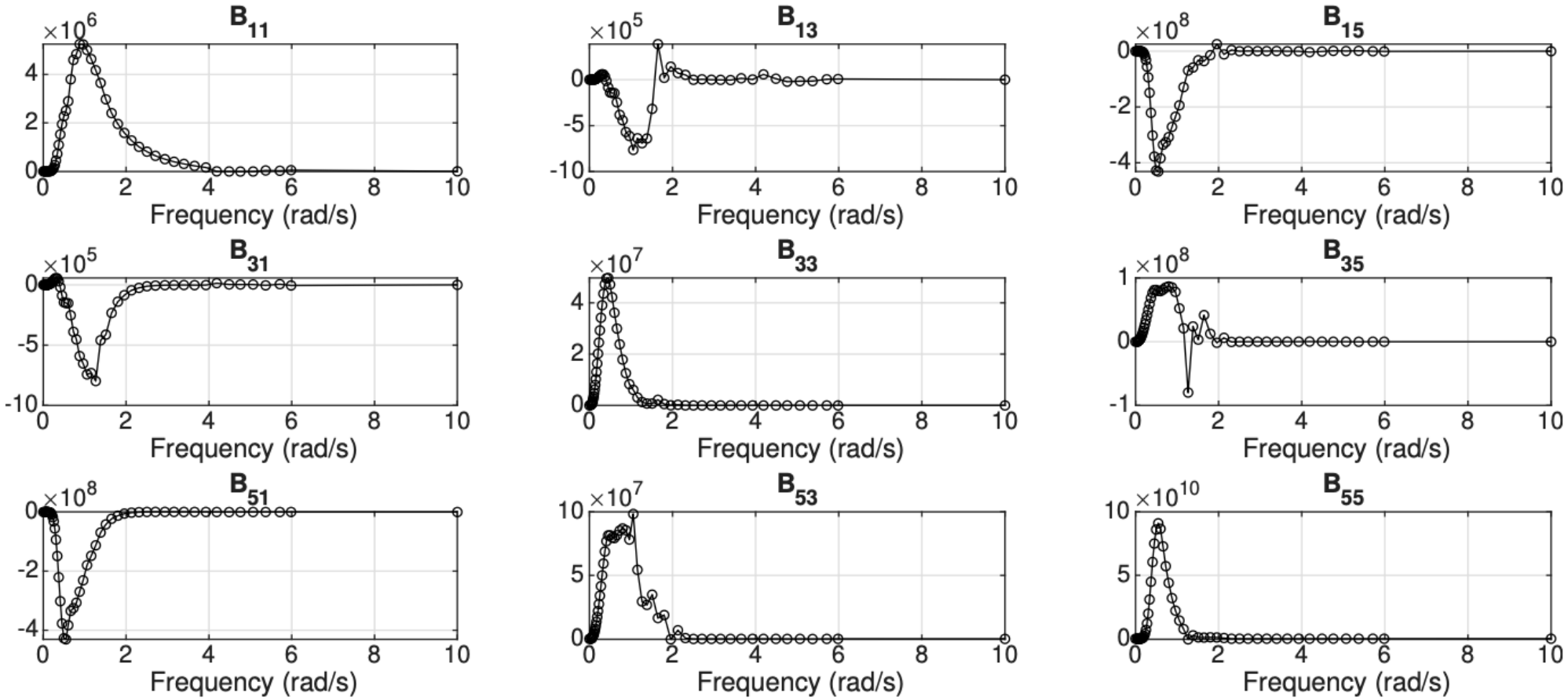
Added-mass coefficients for the MSS FPSO model as a function of frequency, computed using WAMIT. The rightmost point corresponds to the infinite-frequency limit $\omega = \infty$.

Lateral Added-Mass Coefficients



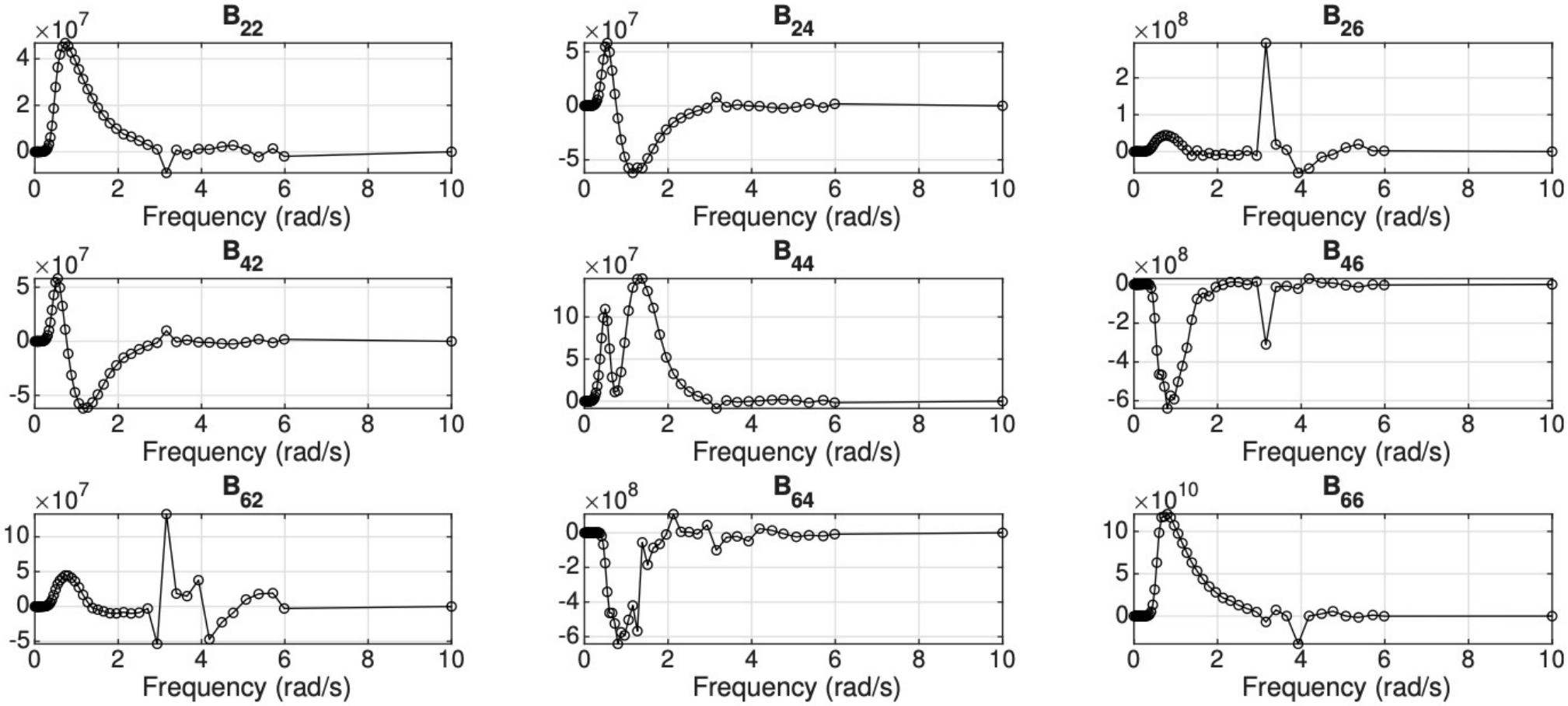
Added-mass coefficients for the MSS FPSO model as a function of frequency, computed using WAMIT. The rightmost point corresponds to the infinite-frequency limit $\omega = \infty$.

Longitudinal Potential-Damping Coefficients



Potential-damping coefficients for the MSS FPSO model as a function of frequency, computed using WAMIT. The rightmost point corresponds to the infinite-frequency limit $\omega = \infty$.

Lateral Potential-Damping Coefficients



Potential-damping coefficients for the MSS FPSO model as a function of frequency, computed using WAMIT. The rightmost point corresponds to the infinite-frequency limit $\omega = \infty$.

Viscous Damping Coefficients

Viscous-Damping Models

The viscous-damping matrix $\boldsymbol{B}_v$ represents hydrodynamic damping effects that are not captured by the potential-flow model. Typical contributions are skin friction, bilge-keel vortices, flow separation, eddy shedding, and other viscous effects. In many applications, $\boldsymbol{B}_v$ is approximated by a diagonal matrix at the CG.

This motivates the following heuristic choice of a diagonal viscous-damping matrix:

$$\boldsymbol{B}_v(\omega) = \operatorname{diag}\{B_{11v} + N_{\mathrm{ITTC}}(A_1),\ B_{22v} + N_{\mathrm{Y,crossflow}}(A_2),\ B_{33v},\ B_{44v},\ B_{55v},\ B_{66v} + N_{\mathrm{N,crossflow}}(A_6)\}$$

- $N_{ITTC}(A_1)$ is an equivalent linear damping coefficient in surge obtained from the ITTC resistance formulation.
- $N_{Y,crossflow}(A_2)$ and $N_{N,crossflow}(A_6)$ are linearized crossflow drag.
- $B_{v,44}$ may be estimated by the empirical method of Ikeda et al. (1976) or by other roll-damping models.

Ikeda, Y., K. Komatsu, Y. Himeno and N. Tanaka (1976). On Roll Damping Force of Ship: Effects of Friction of Hull and Normal Force of Bilge Keels. *Journal of the Kansai Society of Naval Architects* 142, 54–66.

Rule-of-Thumb Specification of Viscous Damping Coefficients

For surge, sway, and yaw, the viscous damping coefficients are specified through desired time constants $T_1^\star$, $T_2^\star$, and $T_6^\star$ for the corresponding first-order mass–damper systems. Using the zero-frequency added-mass coefficients is appropriate because these motions are typically dominated by low-frequency maneuvering dynamics:

$$T_1^\star = \frac{m + A_{11}(0)}{B_{v,11}}, \qquad T_2^\star = \frac{m + A_{22}(0)}{B_{v,22}}, \qquad T_6^\star = \frac{I_z + A_{66}(0)}{B_{v,66}}$$

$$B_{v,11} = \frac{m + A_{11}(0)}{T_1^\star}, \quad B_{v,22} = \frac{m + A_{22}(0)}{T_2^\star}, \quad B_{v,66} = \frac{I_z + A_{66}(0)}{T_6^\star}.$$

It is common to use the zero-frequency added mass values in surge, sway, and yaw, and natural frequencies in heave, roll, and pitch.

We will later learn that there is a magic frequency that can be chosen for 6-DOF coupled motions preserving the kinetic energy and power of the system.

For heave, roll, and pitch, the viscous damping coefficients are specified through desired relative damping ratios $\zeta_3^\star$, $\zeta_4^\star$, and $\zeta_5^\star$ at the natural frequencies ω_3, ω_4, and ω_5. These natural frequencies are obtained from the frequency-dependent added-mass and restoring matrices by solving the generalized eigenvalue problem described in Section 4.3. The viscous damping coefficients $B_{v,33}$, $B_{v,44}$, and $B_{v,55}$ are then computed such that the total damping ratios become equal to the prescribed values,

$$\begin{aligned} B_{v,33} &= 2\zeta_3^\star \omega_3 \left(m + A_{33}(\omega_3)\right) - B_{33}(\omega_3) \\ B_{v,44} &= 2\zeta_4^\star \omega_4 \left(I_x + A_{44}(\omega_4)\right) - B_{44}(\omega_4) \\ B_{v,55} &= 2\zeta_5^\star \omega_5 \left(I_y + A_{55}(\omega_5)\right) - B_{55}(\omega_5). \end{aligned}$$

Alternatively, B_{44v} can be replaced by

$$B_{v,44}(\omega) = \beta_{\text{Ikeda}}(\omega)$$

Linearized Quadratic Surge Resistance $N_{ITTC}(A_1)$

Quadratic surge resistance can be approximated using a "Describing Function"
This is also referred to as "the equivalent linearization method"

$y \approx N(A)x$ Sinusoidal input $x = A\sin(\omega t)$

For static linearities, displaying no dependence upon the derivatives, the describing function for the odd polynomial nonlinearity

$$y = c_1 x + c_2 x|x| + c_3 x^3$$

is

$$N(A) = c_1 + \frac{8A}{3\pi}c_2 + \frac{3A^2}{4}c_3$$

ITTC damping approximation

$$X = X_{|u|u}|u|u \approx -N_{\mathrm{ITTC}}(A_1)u \qquad X_{|u|u} < 0.$$

$$N_{\mathrm{ITTC}}(A_1) = -\frac{8A_1}{3\pi}X_{|u|u}$$

This assumes that the surge velocity is harmonic with amplitude A_1, i.e.

$u = A_1\cos(\omega t)$

Quadratic surge resistance is approximated by replacing the nonlinear quadratic term with an equivalent linear damping coefficient, chosen so that the linearized system produces the same average energy dissipation as the original nonlinear system at a given wave oscillation amplitude.

For a ship moving at high speed, the amplitudes A_2 and A_6 in sway and yaw much smaller than the amplitude in surge A_1. Hence, it is common to neglect these terms in seakeeping analysis.

Gelb, A. and W. E. Vander Velde (1968). Multiple-Input Describing Functions and Nonlinear System Design. McGraw-Hill Book Company.

Linearized Cross-Flow Drag $N_{Y,crossflow}(A_2)$ and $N_{N,crossflow}(A_6)$

Similarly, cross-flow drag in sway and yaw gives

$$Y = Y_{|v|v}|v|v \approx -N_{Y,\text{crossflow}}(A_2)v$$
$$N = N_{|r|r}|r|r \approx -N_{\text{N,crossflow}}(A_6)r$$

$Y_{|v|v} < 0$ and $N_{|r|r} < 0$

$$N_{Y,\text{crossflow}}(A_2) = -\frac{8A_2}{3\pi}Y_{|v|v}$$
$$N_{N,\text{crossflow}}(A_6) = -\frac{8A_6}{3\pi}N_{|r|r}$$

In many seakeeping applications, the amplitudes A_2 and A_6 are smaller than A_1, so the sway- and yaw-damping contributions may be neglected in a first approximation. This assumption should be reconsidered when lateral motions become significant.

Hydrodynamic Programs

Hydrodynamic Concepts and Potential Theory

Potential theory programs typically compute:

- Frequency-dependent added mass matrix, $\boldsymbol{A}(\omega)$
- Frequency-dependent potential damping matrix, $\boldsymbol{B}(\omega)$
- Constant restoring matrix, $\boldsymbol{C}$
- 1st- and 2nd-order wave-induced forces and motions (amplitudes and phases) for given wave directions and frequencies
- … and much more

However, potential theory codes give you no viscous damping terms

$$\left[\boldsymbol{M}_{RB} + \boldsymbol{A}(\omega)\right]\ddot{\boldsymbol{\xi}} + \boldsymbol{B}_{\text{total}}(\omega)\dot{\boldsymbol{\xi}} + \boldsymbol{C}\boldsymbol{\xi} = \boldsymbol{\tau}_{wave}$$

$$\boldsymbol{B}_{\text{total}}(\omega) = \boldsymbol{B}(\omega) + \boldsymbol{B}_v$$

You should add viscous terms manually, if not computed.

One special feature of WAMIT is that the program solves a boundary value problem for zero and infinite added mass. These boundary values are useful when computing the retardation functions in Cummins equation (fluid memory effects).

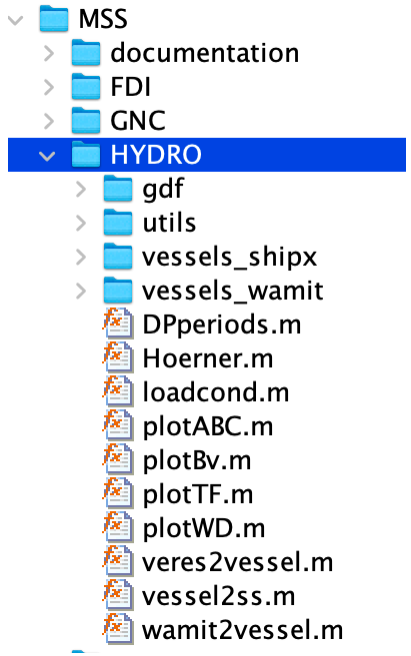
Processing of hydrodynamic data

MSS Toolbox: https://github.com/cybergalactic/MSS

The toolbox reads ASCII data files generated by the hydrodynamic programs:

- **ShipX** – www.sintef.no/en/software/shipx/
- **WAMIT** – www.wamit.com

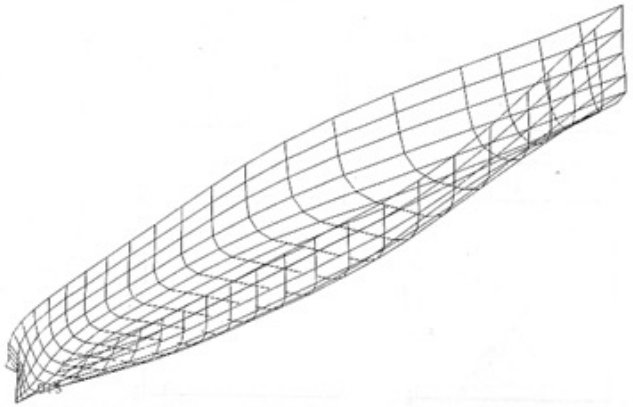
and processes the data for use in Matlab and GNU Octave.

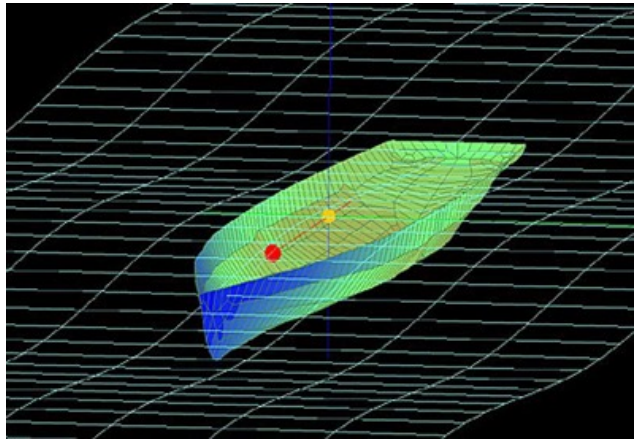


Hydrodynamic Programs

Strip Theory (2-D Potential Theory)

For slender bodies, the motion of the fluid can be formulated as a 2-D problem. An accurate estimate of the hydrodynamic forces can be obtained by applying strip theory (*Newman 1977; Faltinsen 1990*).



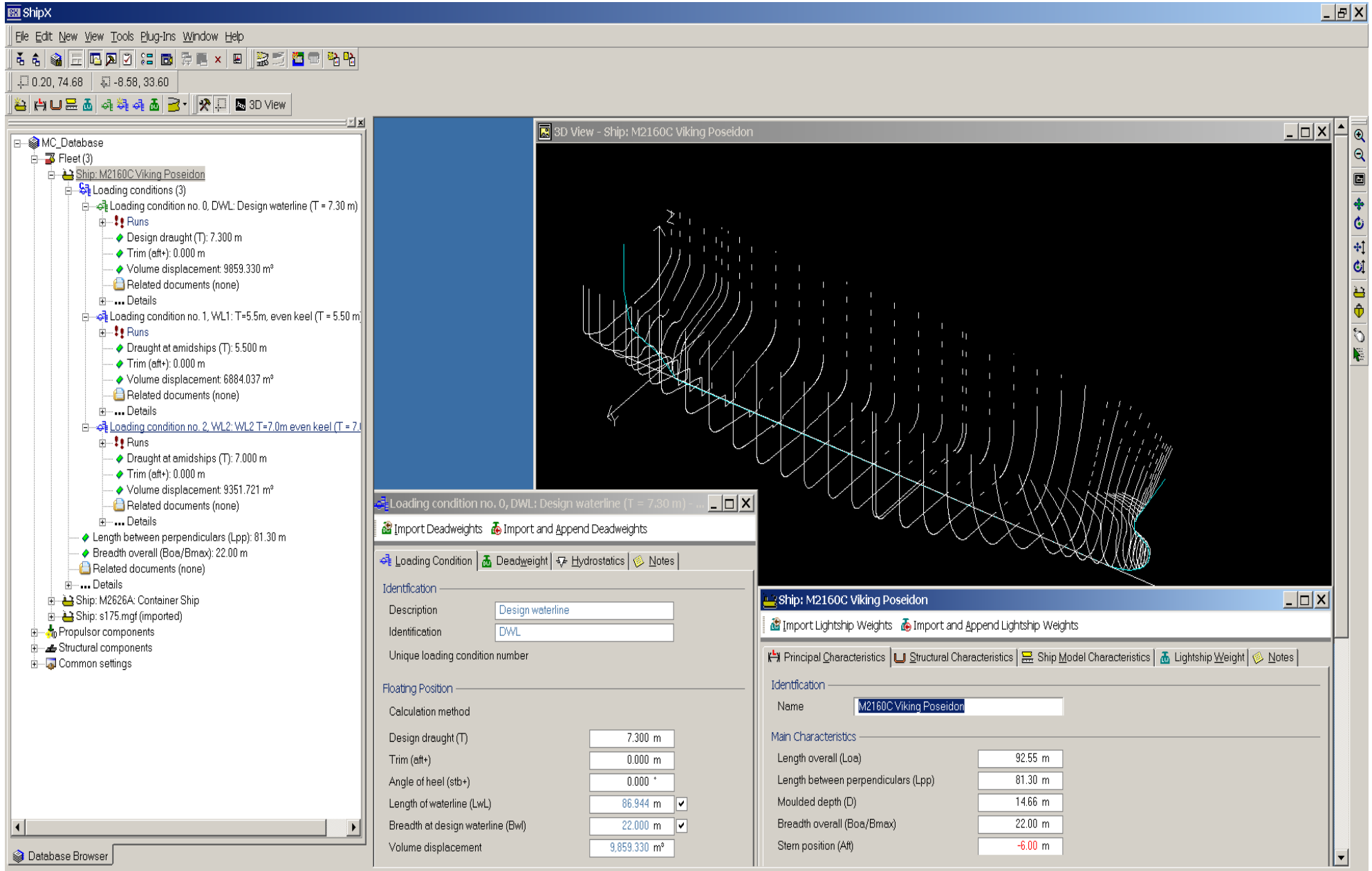


The 2-D theory takes into account that variation of the flow in the cross-directional plane is much larger than the variation in the longitudinal direction of the ship.

The principle of strip theory involves dividing the submerged part of the craft into a finite number of strips. Hence, 2-D hydrodynamic coefficients for added mass can be computed for each strip and then summed over the length of the body to yield the 3-D coefficients.

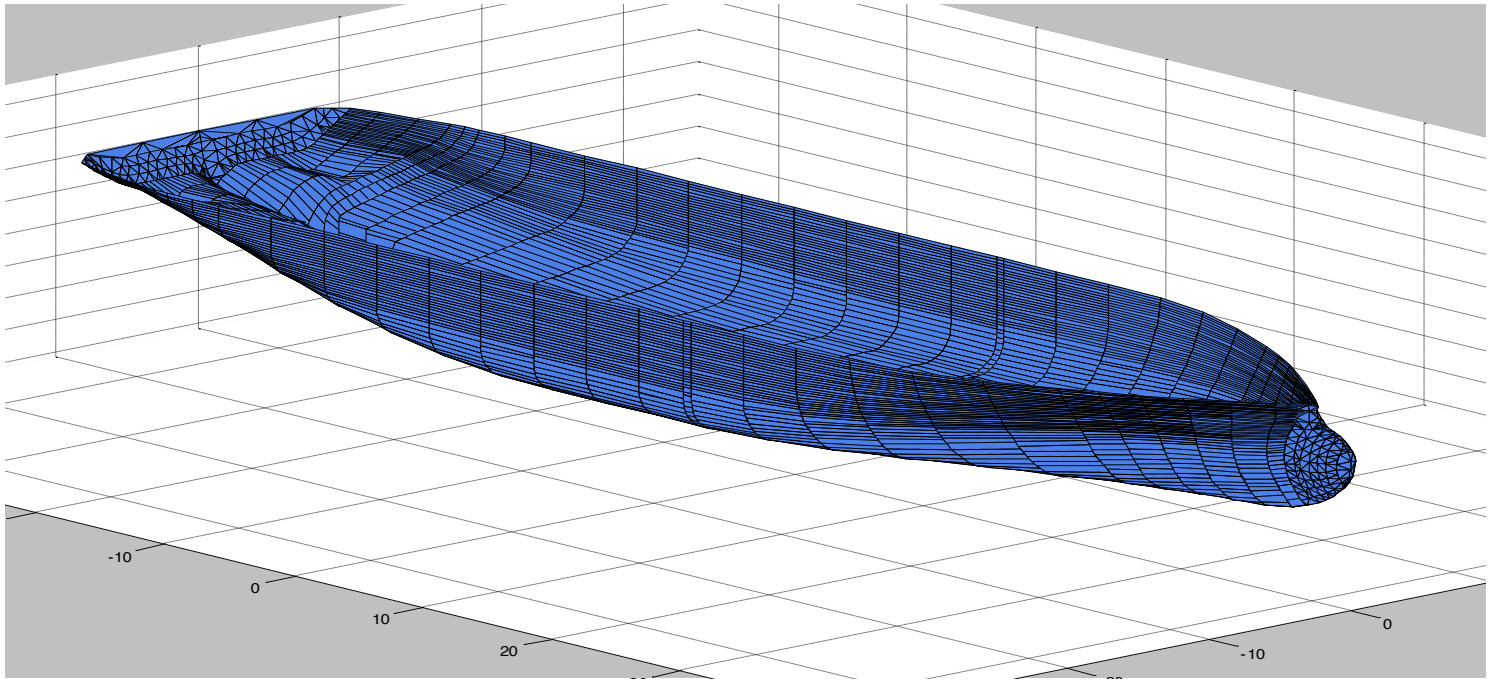
There exists more than 20 commercial strip theory programs and several opens source university codes.

ShipX (VERES) - VEssel RESponse program is a Strip Theory Program which calculates wave-induced loads on and motions of mono-hulls and barges in deep to very shallow water. The program is based on the famous paper by Salvesen, Tuck and Faltinsen (1970). “Ship Motions and Sea Loads”. Transactions of SNAME.



Panel Methods (3-D Potential Theory)

For potential flows, the integrals over the fluid domain can be transformed to integrals over the boundaries of the fluid domain. This allows the application of panel or boundary element methods to solve the 3-D potential theory problem.



3-D panelization of a supply vessel

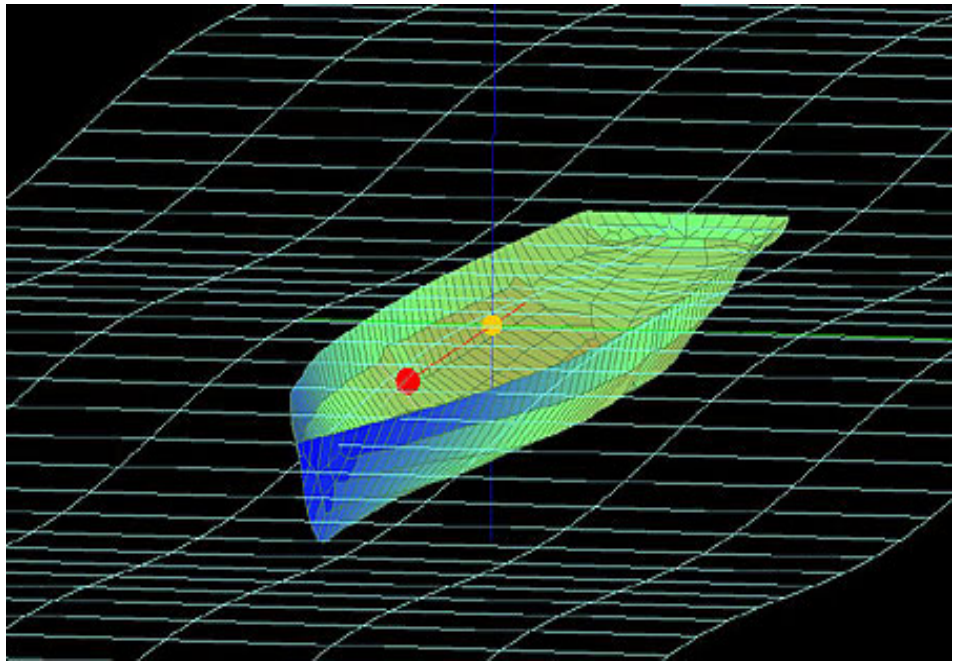
Panel methods divide the surface of the ship and the surrounding water into discrete elements (panels). On each of these elements, a distribution of sources and sinks is defined which fulfill the Laplace equation.

Commercial code: WAMIT (www.wamit.com)

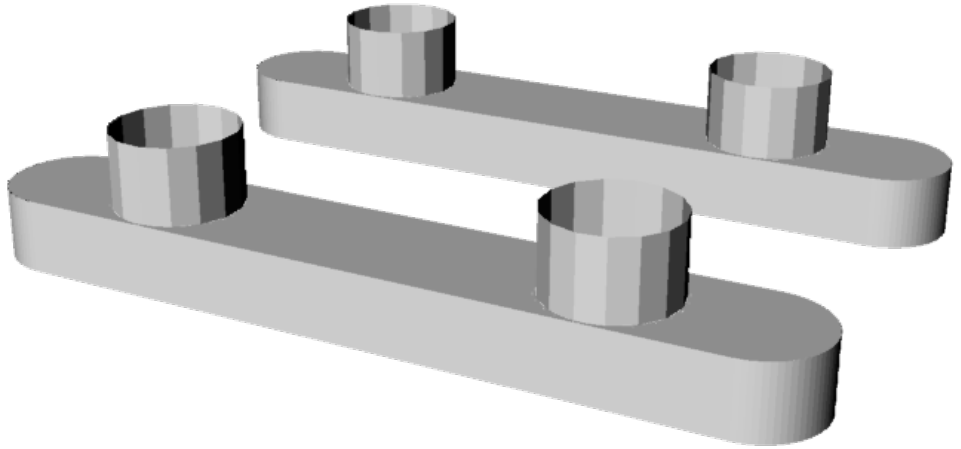
WAMIT

WAMIT® includes advanced tools for analyzing wave interactions with offshore platforms and other structures or vessels.

WAMIT® was developed by Professor Newman and coworkers at MIT in 1987, and it has gained widespread recognition for its ability to analyze the complex structures with a high degree of accuracy and efficiency.



Panelization of semi-submersible using WAMIT user supplied tools



MSS Quick Reference

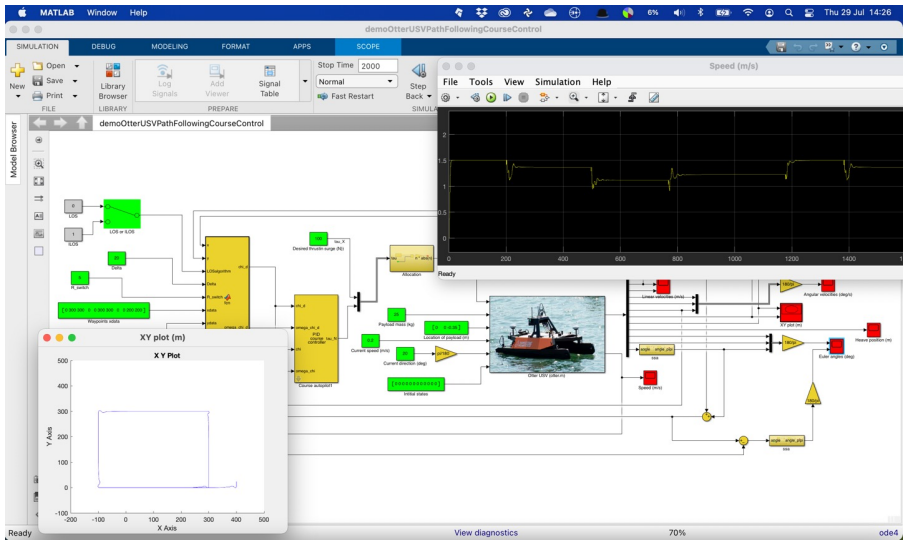
The m-files of the MSS (Marine Systems Simulator) toolbox are compatible with MATLAB (www.mathworks.com) and the free software GNU Octave (www.octave.org), facilitating broad accessibility and application in marine systems simulation. To use the MSS toolbox, please ensure the files are downloaded and correctly set up in your MATLAB/Octave environment. See the guide "How to install MSS for Matlab/Octave?".

```
>> mssHelp        % MSS Quick Reference
>> mssPath        % Build/update and save the MSS path (not available in GNU Octave)
>> mssSimulink    % Simulink Library (not available in GNU Octave)
```

Wamit and ShipX templates

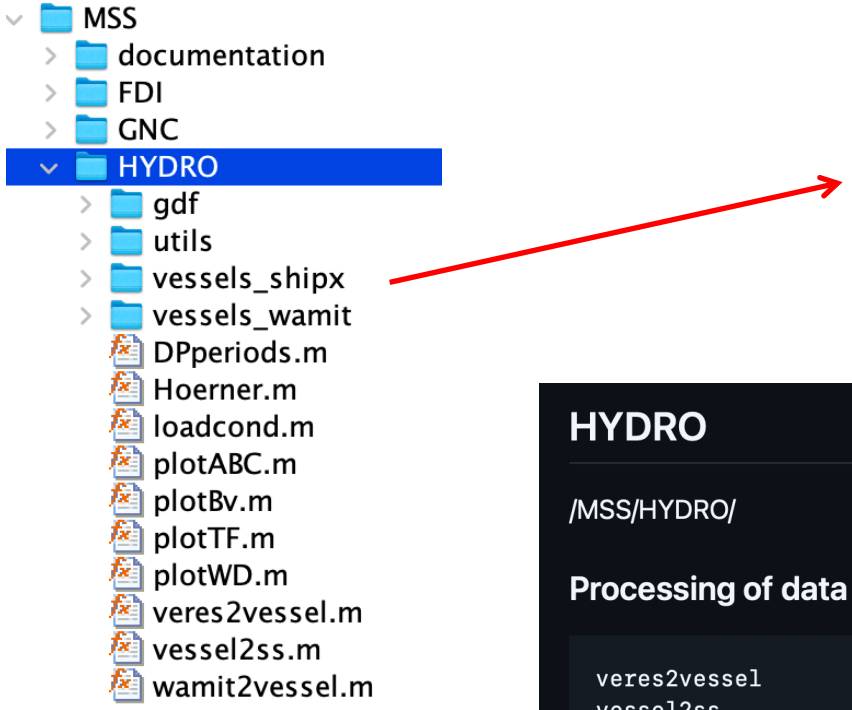
.../MSS/SIMULINK/mssWamitShipxTemplates/

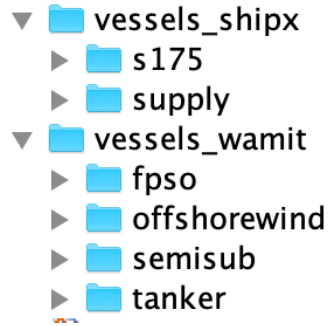
```
DP_ForceRAO.slx    % Simulink template for a DP vessel where wave loads are computed using force RAOs
DP_MotionRAO.slx   % Simulink template for a DP vessel where wave loads are computed using motion RAOs
MAN_ForceRAO.slx   % Simulink template for the unified maneuvering model where wave loads are computed using forc
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MSS Hydro

ShipX and WAMIT data sets in the toolbox





HYDRO

/MSS/HYDRO/

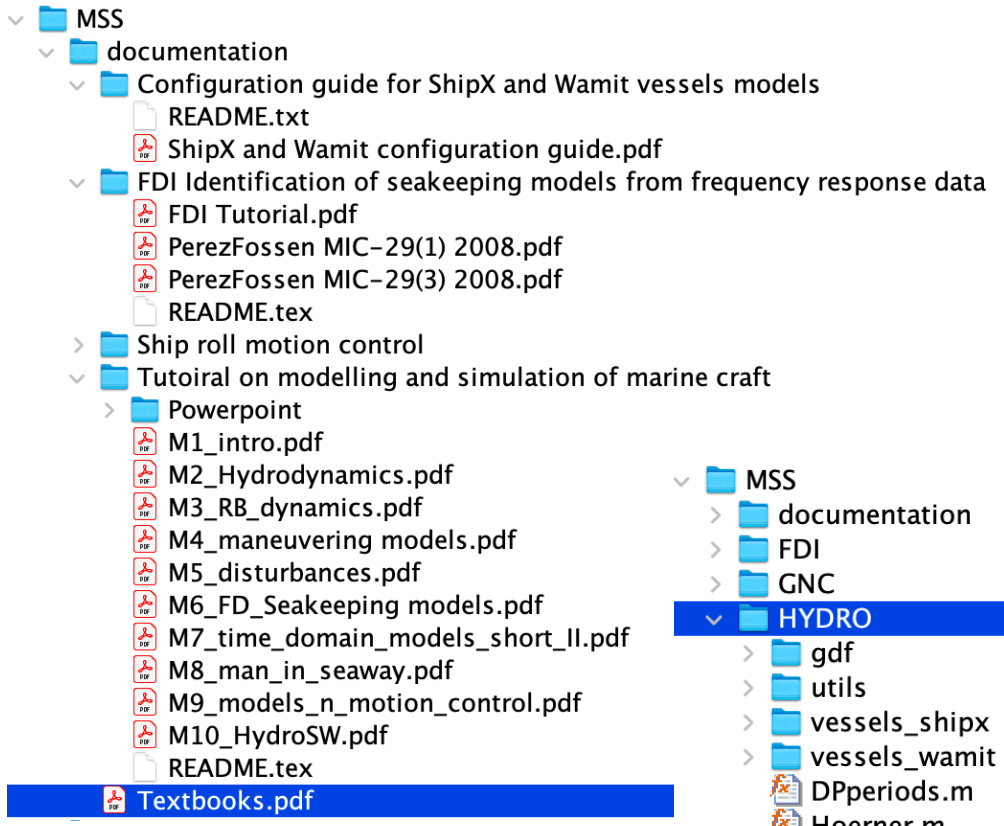
Processing of data from hydrodynamic codes (m-files)

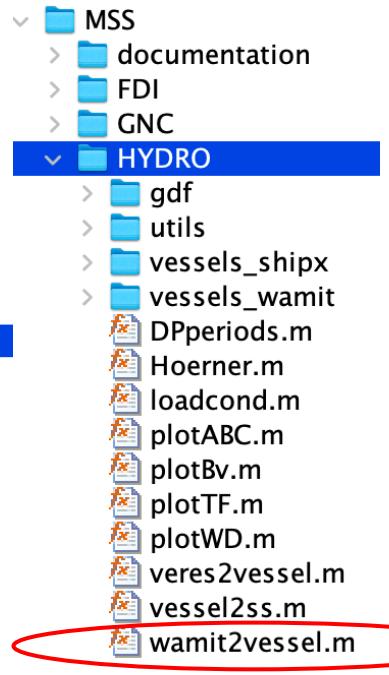
```
veres2vessel      % Reads data from ShipX output files and stores the data as a mat-file containing the structur
vessel2ss         % computes the fluid-memory transfer functions and stores the data as a mat-file containing th
wamit2vessel      % Reads data from WAMIT output files and stores the data as a mat-file containing the structur
```

Data files (mat-files that can be loaded to workspace and used by Simulink templates)

```
fpso, fpsoABC       % WAMIT data for a FPSO
semisub, semisubABC % WAMIT data for a semisubmersible
tanker, tankerABC   % WAMIT data for a tanker
s175, s175ABC       % ShipX data for a supply vessel
supply, supplyABC   % ShipX data for the S175
```

MSS Hydro





Load WAMIT data into Matlab vessel structure

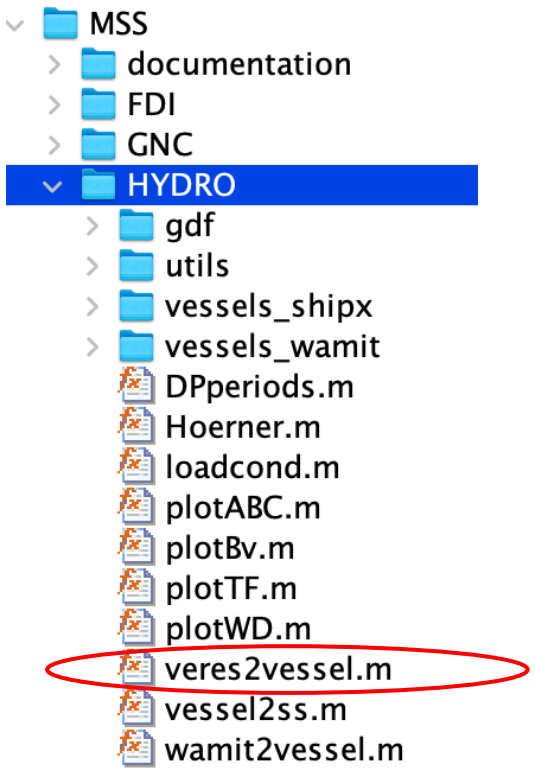
```
vessel = wamit2vessel(filename) reads data from WAMIT output files
and store the data in vessel.mat using the MSS vessel struture. The Wamit
GDF-file must be defined in GLOBAL COORDINATES, i.e. origin [Lpp/2 B/2 WL],
see the Wamit manual. The axes are transformed from Wamit axes to Fossen
(2002) axes.

Example:

>> vessel = wamit2vessel('tanker')
>> vessel = wamit2vessel('tanker',10,246,46)
>> vessel = wamit2vessel('tanker',10,246,46,'1111')

Inputs:
   filename (without extension) reads and processes the following
                WAMIT files:
                   *.1    added mass and damping
                   *.2    force RAOs from Haskind (vessel.forceRAO)
                   *.3    force RAOs from diffraction (vessel.forceRAO)
                   *.4    motion RAOs (vessel.motionRAO)
                   *.8    wave drift data from momentum (vessel.driftfrc)
                   *.9    wave drift data from pressure (vessel.driftfrc)
                   *.frc  rigid body mass parameters
                   *.out  output file
```

MSS Hydro



Load ShipX data into Matlab vessel structure

```
veres2vessel (MSS Hydro)

vessel = veres2vessel(filename, disp_flag) reads data from the
ShipX (Veres) output files *.re1, *.re2, *.re7, *.re8, and *.hyd and
store the data in vesselname.mat using the MSS vessel struture. Examples:

>> veres2vessel('input')
>> veres2vessel('input','1111')

where input are the Veres output data file name.
------------------------------------------------------------------------
Inputs:
  filename: *  (without extension) reads and processes the following
               ShipX (Veres) files:
                  *.re1  motion RAOs (vessel.motionRAO)
                  *.re2  wave drift data (vessel.driftfrc)
                  *.re7  added mass, damping, restoring forces
                  *.re8  force RAOsu (vessel.forceRAO)

   plot_flag (optionally): '1000' plot A and B matrices
                                    '0100' plot force RAOs
                           '0010' plot motion RAOs
                           '0001' plot wave drift forces
                           '0000' NO PLOT
                           '1111' PLOT ALL
------------------------------------------------------------------------
```

MSS Hydro

```
>> load tanker
>> vessel
vessel =
  struct with fields:

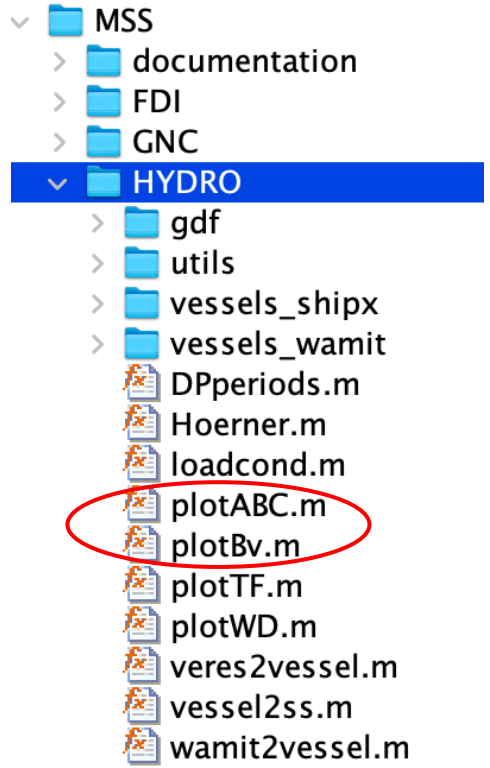
          main: [1×1 struct]
    velocities: 0
      headings: [1×36 double]
           MRB: [6×6 double]
             C: [6×6×60 double]
         freqs: [1×60 double]
             A: [6×6×60 double]
             B: [6×6×60 double]
     motionRAO: [1×1 struct]
      forceRAO: [1×1 struct]
      driftfrc: [1×1 struct]
            Bv: [6×6 double]
```

Viscous damping is added manually (see Section 5.3)

```
>> vessel.main
ans =
  struct with fields:

     name: 'tanker'
        T: 10
        B: 46
      Lpp: 246
        m: 94620210
      rho: 1025
      k44: 17.0200
      k55: 63.9600
      k66: 66.4200
        g: 9.8066
    nabla: 9.2312e+04
       CB: [3.9259 0 5.1845]
     GM_T: 9.9823
     GM_L: 470.9226
       CG: [3.9300 0 12.5000]
>> vessel.A(:,:,1:2)
ans(:,:,1) =
   1.0e+11 *
    0.0000         0   -0.0000         0   -0.0128         0
         0    0.0005         0    0.0030         0    0.0029
   -0.0000         0    0.0039         0    0.0096         0
         0    0.0030         0    0.1103         0   -0.0419
   -0.0128         0    0.0096         0    9.0021         0
         0    0.0029         0   -0.0420         0    1.7383
ans(:,:,2) =
   1.0e+11 *
    0.0000         0   -0.0000         0   -0.0128         0
         0    0.0005         0    0.0030         0    0.0029
   -0.0000         0    0.0040         0    0.0099         0
         0    0.0030         0    0.1103         0   -0.0419
   -0.0128         0    0.0099         0    9.0263         0
         0    0.0029         0   -0.0420         0    1.7407
```

MSS Hydro



Plots the hydrodynamic matrices ***A, B, C*** and B_v as a function of frequency.

```
>> load tanker
>> vessel
vessel =
  struct with fields:

        main: [1×1 struct]
  velocities: 0
    headings: [1×36 double]
         MRB: [6×6 double]
           C: [6×6×60 double]
       freqs: [1×60 double]
           A: [6×6×60 double]
           B: [6×6×60 double]
   motionRAO: [1×1 struct]
    forceRAO: [1×1 struct]
    driftfrc: [1×1 struct]
          Bv: [6×6 double]
```

```
>> plotABC(vessel,mtrx) plots all elements Aij,Bij,Cij versus
   frequency and speed

Inputs:
  vessel:  MSS vessel structure generated from a data file: *
  (file name without extension) by using the following calls:

  ShipX (VERES), MARINTEK :  vessel = veres2vessel(*)
  WAMIT                   :  vessel = wamit2vessel(*)

  mtrx = 'A'    added mass
         'B'    potential + viscous damping
         'C'    restoring forces

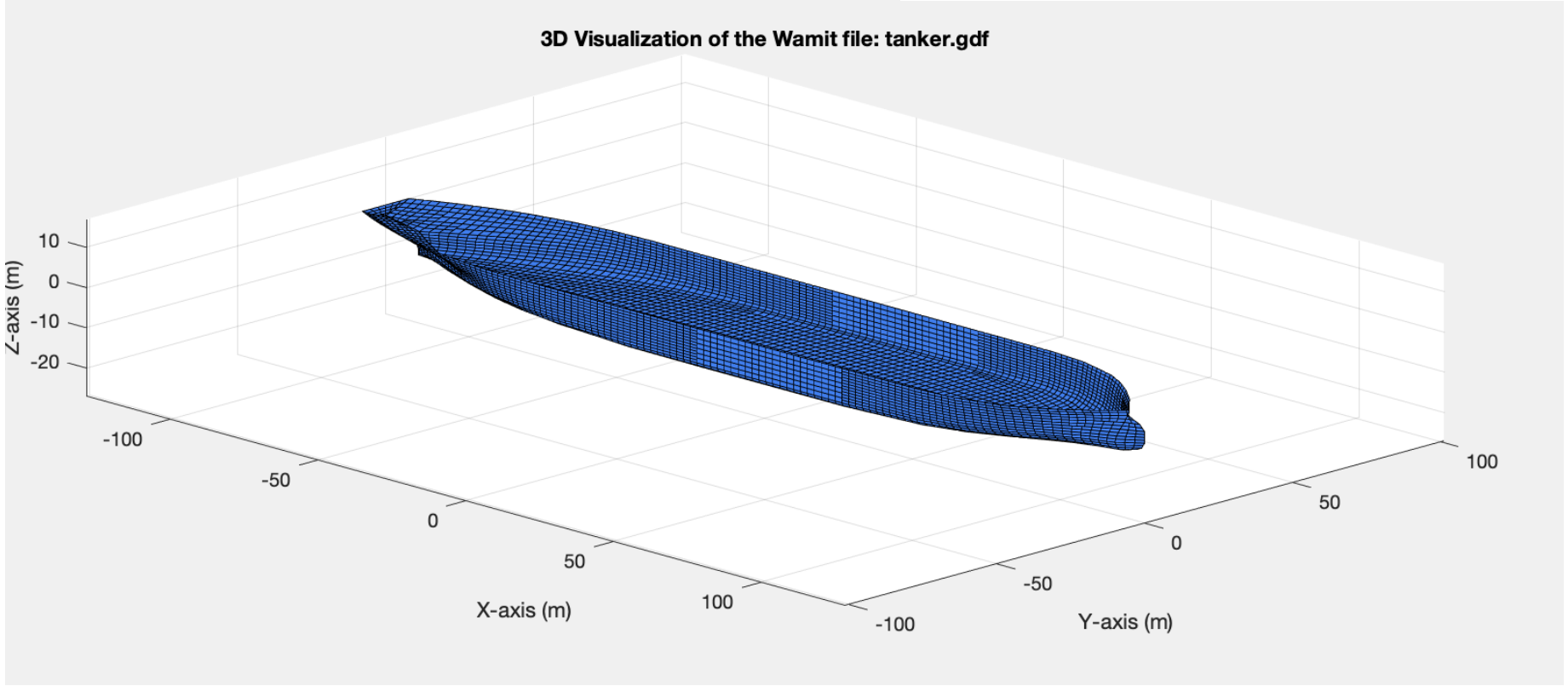
 i, j (optionally):    matrix element
 velno (optionally):   speed number
```

MSS Hydro

```
>> plot_wamitgdf('tanker')
```

```
>> plot_wamitgdf(filename,colorcode,figno,1);
-------------------------------------------------------------------------
Inputs:
   filename:    Low-order Wamit GDF file without extension
   colorcode:   'r','g','b','c','m','y','w','k' or colormaps [0 0 1] etc.
   figno:       figure number
   waterline:   optional flag used to plot free surface points
   half:        optional, use 1 to plot GDF data (no mirror)
                          for Veres data. Octopus data always has mirror

Output:
   gdf_data:    Table dimension N x 12 of Wamit panels where N is the
                number of panels given by 3x4=12 points on each line
```

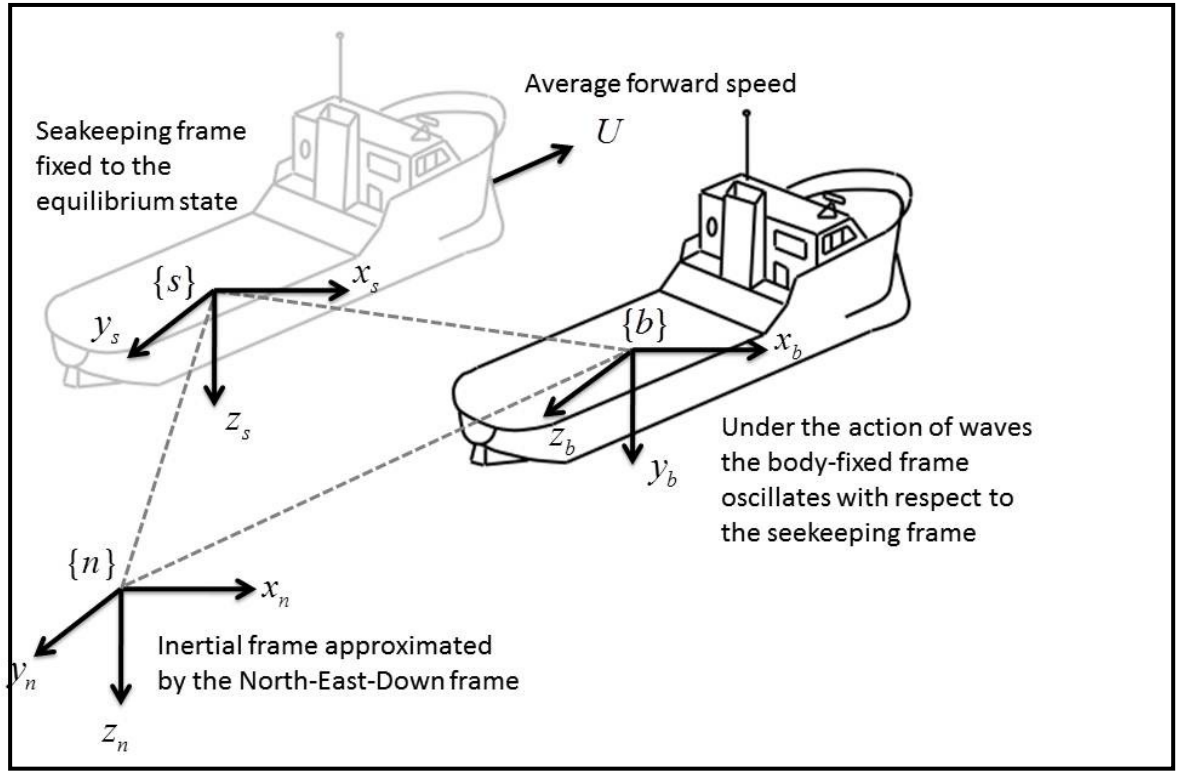


Time-Domain Simulation using Cummins' Equation to Avoid Frequency–Time Mixing

Cummins, W. E. (1962). The Impulse Response Function and Ship Motions. Techn. Report 1661. *David Taylor Model Basin. Hydrodynamics Laboratory*, USA.

Cummins Equation

The seakeeping reference frame {s} is treated as an approximately inertial reference frame.



Equations of Motion Expressed in {s}

$$\boldsymbol{M}_{RB}\ddot{\boldsymbol{\xi}} = \sum_i \boldsymbol{\tau}_i$$

$$\boldsymbol{\xi} = [x^n - \bar{x}^n,\ y^n - \bar{y}^n,\ z^n - \bar{z}^n,\ \phi,\ \theta,\ \psi - \bar{\psi}]^\top$$

Cummins (1962) showed that the radiation-induced hydrodynamic forces in an ideal fluid can be related to frequency-dependent added mass ***A***(ω) and potential damping ***B***(ω).

$$\big(\boldsymbol{M}_{RB} + \boldsymbol{A}(\infty)\big)\ddot{\boldsymbol{\xi}} + \int_0^t \bar{\boldsymbol{K}}(t-\tau)\,\dot{\boldsymbol{\xi}}(\tau)\,d\tau + \boldsymbol{C}\boldsymbol{\xi} = \boldsymbol{\tau}_{wave}$$

$$\boldsymbol{\tau}_{\text{hyd}} = -\bar{\boldsymbol{A}}\ddot{\boldsymbol{\xi}} - \int_0^t \bar{\boldsymbol{K}}(t-\tau)\dot{\boldsymbol{\xi}}(\tau)\mathrm{d}\tau$$

$$\bar{\boldsymbol{A}} = \boldsymbol{A}(\infty)$$

$$\bar{\boldsymbol{K}}(t) = \frac{2}{\pi}\int_0^\infty \boldsymbol{B}(\omega)\cos(\omega t)\mathrm{d}\omega$$

Cummins, W. E. (1962). The Impulse Response Function and Ship Motions. Technical Report 1661. David Taylor Model Basin. Hydrodynamics Laboratory, USA.

Inconsistent Mixing of Time- and Frequency-Domain Hydrodynamic Models

$$(\boldsymbol{M}_{RB}+\boldsymbol{A}(\omega))\,\ddot{\boldsymbol{\xi}}+(\boldsymbol{B}(\omega)+\boldsymbol{B}_v)\,\dot{\boldsymbol{\xi}}+\boldsymbol{C}\boldsymbol{\xi}=\delta\boldsymbol{\tau}+\boldsymbol{\tau}_{\text{wind}}+\boldsymbol{\tau}_{\text{wave}}+\boldsymbol{\tau}_{\text{current}}$$

This model is well established in the hydrodynamics literature. However, the commonly used "time-domain" form is not a true ODE because it mixes time- and frequency-domain quantities through an abuse of notation. Hence, the model is not an ODE depending only on time, but an integro-differential equation.

The correct alternative is Cummins' (1962) time-domain model, which is mathematically consistent with the frequency-domain formulation and therefore captures vessel responses over the full frequency range:

$$(\boldsymbol{M}_{RB}+\boldsymbol{A}(\infty))\,\ddot{\boldsymbol{\xi}}+(\boldsymbol{B}(\infty)+\boldsymbol{B}_v)\,\dot{\boldsymbol{\xi}}+\int_0^t \boldsymbol{K}(t-\tau)\dot{\boldsymbol{\xi}}(\tau)\mathrm{d}\tau+\boldsymbol{C}\boldsymbol{\xi}=\delta\boldsymbol{\tau}+\boldsymbol{\tau}_{\text{wind}}+\boldsymbol{\tau}_{\text{wave}}+\boldsymbol{\tau}_{\text{current}}$$

$$\boldsymbol{K}(t)=\frac{2}{\pi}\int_0^\infty [\boldsymbol{B}(\omega)-\boldsymbol{B}(\infty)]\cos(\omega t)\,\mathrm{d}\omega$$

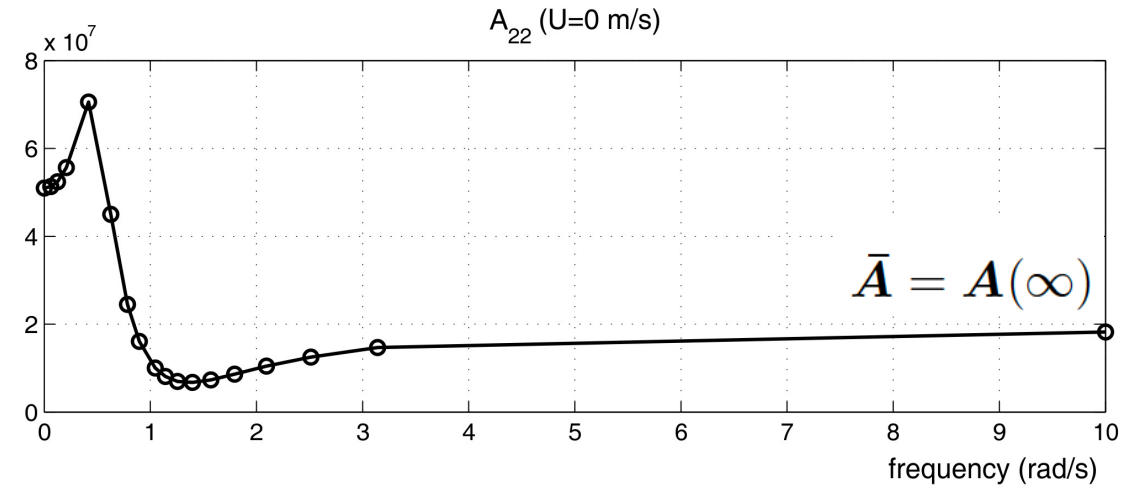
Ogilvie (1964) has shown that from a numerical point of view it is better to integrate the difference $\boldsymbol{B}(\omega)$-$\boldsymbol{B}(\infty)$ than $\boldsymbol{B}(\omega)$

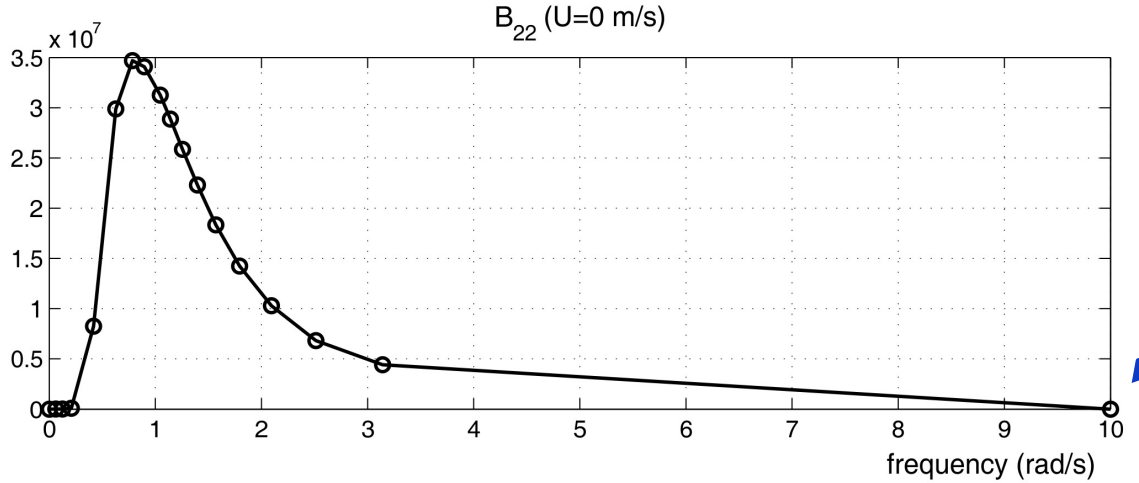
If the matrices $\boldsymbol{A}(\omega)$ and $\boldsymbol{B}(\omega)$ are computed at one constant frequency, the hydrodynamic mass–damper–spring system reduces to an ODE with constant coefficients. This system can then be solved using standard numerical integration methods. We will now discuss approaches for this such as the zero-frequency method and the power-based averaging method.

Cummins, W. E. (1962). The Impulse Response Function and Ship Motions. Techn. Report 1661. David Taylor Model Basin. Hydrodynamics Laboratory, USA.
Ogilvie, T. F. (1964). Recent Progress Towards the Understanding and Prediction of Ship Motions. 5th Symposium on Naval Hydrodynamics, pp. 3–79.

Frequency-Dependent Potential Coefficients

Frequency-dependent added mass $A_{22}(\omega)$ and potential damping $B_{22}(\omega)$ in sway





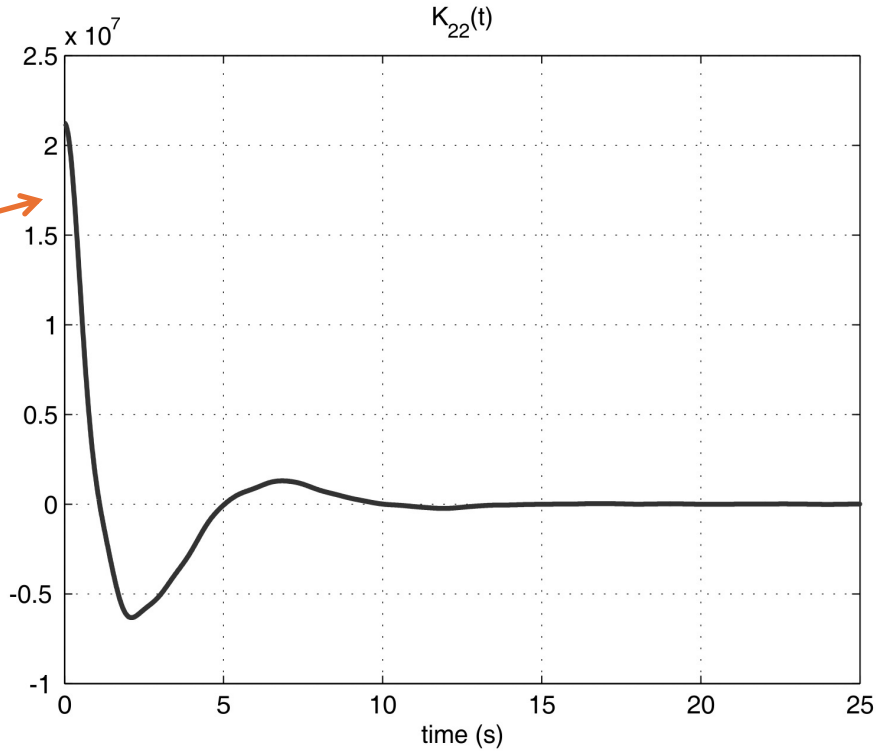
Infiinity added mass and damping
$A_{22}(\infty)$ and $B_{22}(\infty)$

Retardation Functions and Fluid Memory Effects

$$\boldsymbol{\tau}_{\text{hyd}} = -\bar{\boldsymbol{A}}\ddot{\boldsymbol{\xi}} - \int_0^t \bar{\boldsymbol{K}}(t-\tau)\dot{\boldsymbol{\xi}}(\tau)\mathrm{d}\tau$$

Matrix of retardation functions

$$\boldsymbol{K}(t) = \frac{2}{\pi}\int_0^\infty [\boldsymbol{B}(\omega) - \boldsymbol{B}(\infty)]\cos(\omega t)\,\mathrm{d}\omega$$



Cummins Equation

If linear restoring forces $\boldsymbol{\tau}_{hs} = -\boldsymbol{C\xi}$ are included in the model, this results in the time-domain model

$$\left(\boldsymbol{M}_{RB} + \boldsymbol{A}(\infty)\right)\ddot{\boldsymbol{\xi}} + \int_0^t \bar{\boldsymbol{K}}(t-\tau)\,\dot{\boldsymbol{\xi}}(\tau)\,d\tau + \boldsymbol{C\xi} = \boldsymbol{\tau}_{wave}$$

The fluid memory effects can be replaced by a state-space model to avoid the integral. This approach typically results in 3-5 states for each matrix element. Hence, you might need to solve 36 x 5 ODEs in real time.

In mathematics, an integro-differential equation is an equation that involves both integrals and derivatives of a function.

Cummins' Equation in BODY Coordinates

$$(\boldsymbol{M}_{RB} + \boldsymbol{A}(\infty))\,\ddot{\boldsymbol{\xi}} + (\boldsymbol{B}(\infty) + \boldsymbol{B}_v)\,\dot{\boldsymbol{\xi}} + \int_0^t \boldsymbol{K}(t-\tau)\dot{\boldsymbol{\xi}}(\tau)\mathrm{d}\tau + \boldsymbol{C}\boldsymbol{\xi} = \delta\boldsymbol{\tau} + \boldsymbol{\tau}_{\text{wind}} + \boldsymbol{\tau}_{\text{wave}} + \boldsymbol{\tau}_{\text{current}}$$

The SEAKEEPING model is transformed from {s} to {b} using the perturbation coordinates

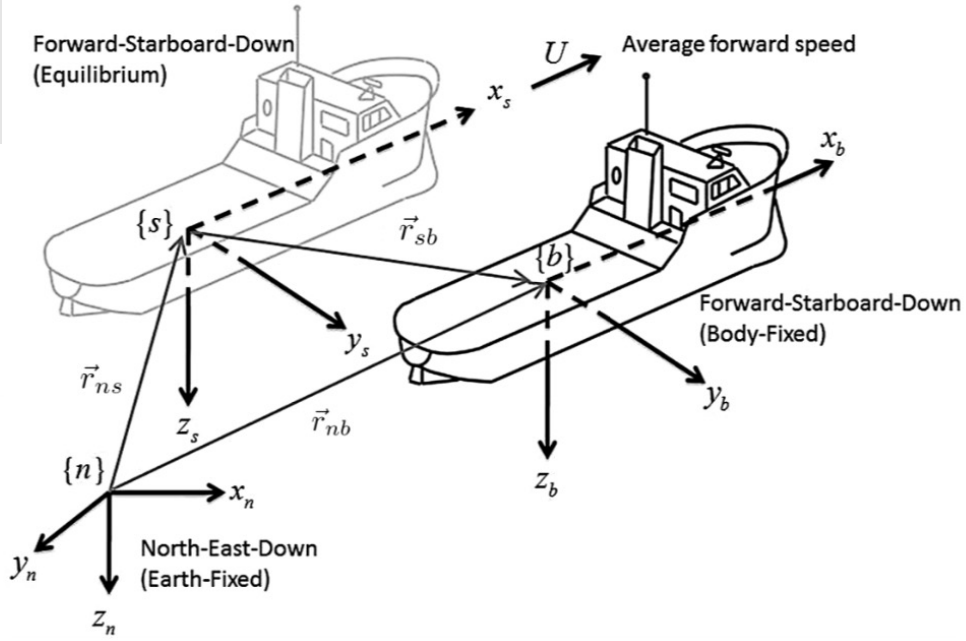
$$\boldsymbol{\xi} := \delta\boldsymbol{\eta} = \boldsymbol{\eta} - \bar{\boldsymbol{\eta}}$$

$$\dot{\boldsymbol{\xi}} = \delta\boldsymbol{\nu}, \qquad \ddot{\boldsymbol{\xi}} = \delta\dot{\boldsymbol{\nu}}$$

$$\delta\boldsymbol{\nu} = \boldsymbol{\nu} - \bar{\boldsymbol{\nu}}, \qquad \delta\boldsymbol{\tau} = \boldsymbol{\tau} - \bar{\boldsymbol{\tau}}.$$

The steady-state control input required to maintain the constant equilibrium velocity in still water is

$$\bar{\boldsymbol{\tau}} = (\boldsymbol{B}(\infty) + \boldsymbol{B}_v)\,\bar{\boldsymbol{\nu}}$$



$$(\boldsymbol{M}_{RB} + \boldsymbol{A}(\infty))\,\dot{\boldsymbol{\nu}} + (\boldsymbol{B}(\infty) + \boldsymbol{B}_v)\,\boldsymbol{\nu} + \int_0^t \boldsymbol{K}(t-\tau)\delta\boldsymbol{\nu}(\tau)\,\mathrm{d}\tau + \boldsymbol{C}\boldsymbol{\xi} = \boldsymbol{\tau} + \boldsymbol{\tau}_{\text{wave}} + \boldsymbol{\tau}_{\text{wind}} + \boldsymbol{\tau}_{\text{current}}$$

$$\boldsymbol{\tau}_{\text{current}} = \boldsymbol{A}(\infty)\dot{\boldsymbol{\nu}}_c + (\boldsymbol{B}(\infty) + \boldsymbol{B}_v)\,\boldsymbol{\nu}_c$$

Time-Domain Simulation using the Power-Based Averaging Method to avoid Zero-Frequency Approximations

Fossen, T. I. (2025). Maneuvering Coefficient Estimation from Frequency-Dependent Added Mass and Damping: A Power-Based Approach. Oceanic Engineering 341, 122494. https://doi.org/10.1016/j.oceaneng.2025.122494

Constant Equivalent Added Mass and Damping Matrices

In principle, the time-domain radiation force can be modeled exactly using Cummins' equation with retardation functions. However, this approach is rarely used in maneuvering and control applications because:

- *The retardation functions are difficult to obtain and implement.*
- *The method is computationally expensive in time-domain simulations.*
- *Numerical stability and accuracy issues arise, especially in long-term simulations.*

To avoid these complexities, we will instead compute constant equivalent added mass and damping matrices $\boldsymbol{A}_{\mathrm{U}}^{\mathrm{eq}}$ and $\boldsymbol{B}_{\mathrm{U}}^{\mathrm{eq}}$ directly from frequency-domain data (Fossen, 2025).

Zero-Frequency Method

- Maneuvering models often use constant matrices from zero-frequency limits:
  → Because surge, sway, and yaw are dominated by low-frequency dynamics, where radiation damping vanishes.
- Fails in 6-DOF coupled systems:
  → Couplings (A_{13}, A_{15}, A_{24}, A_{26}, A_{35}, A_{46}) peak at different frequencies. Consequently, the zero-frequency values are not representative in 6-DOF motions.

Power-Based Averaging Method

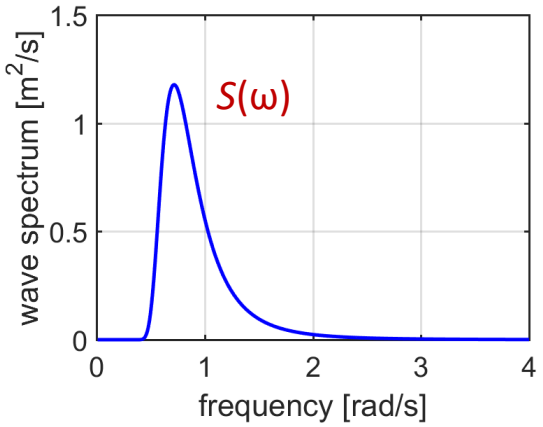
- Uses wave spectral energy density as weighting (justified by how wave forcing distributes energy across frequencies).
- Ensures kinetic energy and mean power dissipation are preserved in the equivalent matrices.

Spectral Averaging for Equivalent Hydrodynamic Coefficients

Pierson–Moskowitz (PM) spectrum

$$S(\omega) = \frac{\alpha g^2}{\omega^5} \exp\left(-\beta \left(\frac{\omega_p}{\omega}\right)^4\right)$$

$\alpha = 8.1\ 10^{-3}$
$\beta = 0.74$
$g = 9.81\ m/s^2$



This formulation is a reparameterization of the classical PM spectrum used for weighting, where ω_p is a tunable parameter representing the spectral peak and ω is the wave frequency.

Normalized wave spectrum

$$S^N(\omega) = \frac{S(\omega)}{\int_0^\infty S(\omega)\,\mathrm{d}\omega}, \qquad \int_0^\infty S^N(\omega)\,\mathrm{d}\omega = 1$$

For vessels advancing with forward speed U > 0, the wave spectrum can be evaluated in the encounter frequency domain using:

$$S_e^N(\omega_e) = \frac{S_e(\omega_e)}{\int_0^\infty S_e(\omega_e)\,\mathrm{d}\omega_e}, \qquad \int_0^\infty S_e^N(\omega_e)\,\mathrm{d}\omega_e = 1.$$

$$S_e(\omega_e) = S(\omega)\left|\frac{\mathrm{d}\omega}{\mathrm{d}\omega_e}\right| \qquad \omega_e = \omega - \frac{\omega^2}{g} U \cos(\beta_{\mathrm{rel}})$$

where $\beta_{rel} = \beta_{wave} - \psi$ is the relative wave direction, ψ is the heading angle and β_{wave} is the wave direction relative to the vessel heading, with 0° for following seas and 180° for head seas.

Equivalent Added Mass and Damping Matrices

Frequency-dependent added mass $\mathbf{A_U}(\omega)$ and potential damping $\mathbf{B_U}(\omega)$ for constant speed U are integrated over the full frequency band relevant to a given sea state, weighted by the spectral energy distribution using a normalized wave spectrum to ensure that only the spectral shape influences the equivalent values $\boldsymbol{A}_{\mathrm{U}}^{\mathrm{eq}}$ and $\boldsymbol{B}_{\mathrm{U}}^{\mathrm{eq}}$ (Fossen, 2025).

$$\boldsymbol{A}_U^{\mathrm{eq}} := \int_0^\infty \boldsymbol{A}_U(\omega_e)\, S_e^N(\omega_e)\, \mathrm{d}\omega_e$$
$$\boldsymbol{B}_U^{\mathrm{eq}} := \int_0^\infty \boldsymbol{B}_U(\omega_e)\, S_e^N(\omega_e)\, \mathrm{d}\omega_e.$$

In practical applications, the zeroth spectral moment m_0 is evaluated for $U = 0$, thereby avoiding explicit dependence of the weighting spectrum on the forward speed U and relative wave heading β_{rel} such that $\omega_e = \omega$. The peak frequency ω_p is then treated as a constant parameter describing the average sea state for the operating condition under consideration.

$$\boldsymbol{A}_U^{\mathrm{eq}} \approx \frac{1}{m_0} \sum_{k=1}^{N} \boldsymbol{A}_U(\omega[k])\, S(\omega[k])\, \Delta\omega$$
$$\boldsymbol{B}_U^{\mathrm{eq}} \approx \frac{1}{m_0} \sum_{k=1}^{N} \boldsymbol{B}_U(\omega[k])\, S(\omega[k])\, \Delta\omega.$$

$$S^N(\omega) = \frac{S(\omega)}{m_0}$$
$$m_0 = \int_0^{\omega_{\max}} S(\omega)\, \mathrm{d}\omega \approx \sum_{k=1}^{N} S(\omega[k])\, \Delta\omega$$

Energy-Consistent Equivalent Added Mass

$$\begin{aligned}\mathbb{E}(K) &= \frac{1}{2}\int_0^\infty \dot{\boldsymbol{\eta}}^\top \boldsymbol{A}_U(\omega)\dot{\boldsymbol{\eta}}\, S_e^N(\omega)\, \mathrm{d}\omega \\ &= \frac{1}{2}\dot{\boldsymbol{\eta}}^\top \left(\int_0^\infty \boldsymbol{A}_U(\omega)\, S_e^N(\omega)\, \mathrm{d}\omega\right)\dot{\boldsymbol{\eta}} \\ &= \frac{1}{2}\dot{\boldsymbol{\eta}}^\top \boldsymbol{A}_U^{\mathrm{eq}}\dot{\boldsymbol{\eta}}\end{aligned}$$

Power-Consistent Equivalent Damping

$$\begin{aligned}\mathbb{E}(P) &= \int_0^\infty \dot{\boldsymbol{\eta}}^\top \boldsymbol{B}_U(\omega)\dot{\boldsymbol{\eta}}\, S_e^N(\omega)\, \mathrm{d}\omega \\ &= \dot{\boldsymbol{\eta}}^\top \left(\int_0^\infty \boldsymbol{B}_U(\omega)\, S_e^N(\omega)\, \mathrm{d}\omega\right)\dot{\boldsymbol{\eta}} \\ &= \dot{\boldsymbol{\eta}}^\top \boldsymbol{B}_U^{\mathrm{eq}}\dot{\boldsymbol{\eta}}\end{aligned}$$

Fossen, T. I. (2025). Maneuvering Coefficient Estimation from Frequency-Dependent Added Mass and Damping: A Power-Based Approach. Oceanic Engineering 341, 122494. https://doi.org/10.1016/j.oceaneng.2025.122494

Time-Domain Seakeeping Model with Equivalent Matrices

Classical Frequency-Domain Seakeeping Model

$$(\boldsymbol{M}_{RB}+\boldsymbol{A}(\omega))\,\ddot{\boldsymbol{\xi}}+(\boldsymbol{B}(\omega)+\boldsymbol{B}_v)\,\dot{\boldsymbol{\xi}}+\boldsymbol{C}\boldsymbol{\xi}=\delta\boldsymbol{\tau}+\boldsymbol{\tau}_{\text{wind}}+\boldsymbol{\tau}_{\text{wave}}+\boldsymbol{\tau}_{\text{current}}$$

Using constant equivalent matrices to avoid frequency dependency

Time-Domain Seakeeping Model with Constant Equivalent Matrices

$$(\boldsymbol{M}_{RB}+\boldsymbol{A}_U^{\text{eq}})\,\ddot{\boldsymbol{\xi}}+(\boldsymbol{B}_U^{\text{eq}}+\boldsymbol{B}_v)\,\dot{\boldsymbol{\xi}}+\boldsymbol{C}\boldsymbol{\xi}=\delta\boldsymbol{\tau}+\boldsymbol{\tau}_{\text{wave}}+\boldsymbol{\tau}_{\text{wind}}+\boldsymbol{\tau}_{\text{current}}$$

The equivalent matrices are computed for a specified operating condition characterized by a constant forward speed U and relative wave heading $\beta_{\text{rel}}=\beta_{\text{wave}}-\psi$ where ψ is the mean vessel heading. In practice, $\boldsymbol{A}_{\text{U}}^{\text{eq}}$ and $\boldsymbol{B}_{\text{U}}^{\text{eq}}$ are computed for the current sea state and then treated as constant matrices in the time-domain simulation until the sea state changes, at which point updated values can be computed efficiently.

In practical implementations, the dependence on the encounter frequency ω_{e} is often neglected when computing the equivalent matrices. Instead, the weighting spectrum is evaluated for $U=0$, and the peak frequency ω_p is chosen as a constant parameter representing the average sea state for the operating condition.

The matrices $\boldsymbol{A}_{\text{U}}^{\text{eq}}$ and $\boldsymbol{B}_{\text{U}}^{\text{eq}}$ and are then obtained directly from $S^N(\omega)$, while the effects of forward speed and relative wave heading are retained in the wave-excitation RAOs through the encounter-frequency mapping. In this way, different operating conditions are represented in practice by selecting different values of ω_p.

Seakeeping Model with Equivalent Matrices in BODY Coordinates

$$(\boldsymbol{M}_{RB} + \boldsymbol{A}_U^{\text{eq}})\,\delta\dot{\boldsymbol{\nu}} + (\boldsymbol{B}_U^{\text{eq}} + \boldsymbol{B}_v)\,\delta\boldsymbol{\nu} + \boldsymbol{C}\boldsymbol{\xi} = (\boldsymbol{\tau} - \bar{\boldsymbol{\tau}}) + \boldsymbol{\tau}_{\text{wave}} + \boldsymbol{\tau}_{\text{wind}} + \boldsymbol{\tau}_{\text{current}}$$

The SEAKEEPING model is transformed from {s} to {b} using the perturbation coordinates

$$\boldsymbol{\xi} := \delta\boldsymbol{\eta} = \boldsymbol{\eta} - \bar{\boldsymbol{\eta}}$$

$$\delta\boldsymbol{\nu} = \boldsymbol{\nu} - \bar{\boldsymbol{\nu}}, \qquad \delta\boldsymbol{\tau} = \boldsymbol{\tau} - \bar{\boldsymbol{\tau}}.$$

$$\dot{\boldsymbol{\xi}} = \delta\boldsymbol{\nu}, \qquad \ddot{\boldsymbol{\xi}} = \delta\dot{\boldsymbol{\nu}}$$

The steady-state control input required to maintain the constant equilibrium velocity in still water is

$$\bar{\boldsymbol{\tau}} = (\boldsymbol{B}_U^{\text{eq}} + \boldsymbol{B}_v)\bar{\boldsymbol{\nu}}$$

$$(\boldsymbol{M}_{RB} + \boldsymbol{A}_U^{\text{eq}})\,\dot{\boldsymbol{\nu}} + (\boldsymbol{B}_U^{\text{eq}} + \boldsymbol{B}_v)\,\boldsymbol{\nu} + \boldsymbol{C}\boldsymbol{\xi} = \boldsymbol{\tau} + \boldsymbol{\tau}_{\text{wave}} + \boldsymbol{\tau}_{\text{wind}} + \boldsymbol{\tau}_{\text{current}}$$

Nonlinear Unified Maneuvering Model using Relative Velocity

$$(\boldsymbol{M}_{RB}+\boldsymbol{A}_U^{\mathrm{eq}})\,\dot{\boldsymbol{\nu}}+(\boldsymbol{B}_U^{\mathrm{eq}}+\boldsymbol{B}_v)\,\boldsymbol{\nu}+\boldsymbol{C}\boldsymbol{\xi}=\boldsymbol{\tau}+\boldsymbol{\tau}_{\mathrm{wave}}+\boldsymbol{\tau}_{\mathrm{wind}}+\boldsymbol{\tau}_{\mathrm{current}}.$$

Maneuvering Model Definitions (Constant Matrices)

$$\boldsymbol{M}_A := \boldsymbol{A}_0^{\mathrm{eq}}, \qquad \boldsymbol{D} := \boldsymbol{B}_0^{\mathrm{eq}}+\boldsymbol{B}_v, \qquad \boldsymbol{G} := \boldsymbol{P}_{345}\boldsymbol{C}\boldsymbol{P}_{345}.$$

$$\boldsymbol{M}=\boldsymbol{M}_{RB}+\boldsymbol{M}_A, \qquad \boldsymbol{C}(\boldsymbol{\nu}_r)=\boldsymbol{C}_{RB}(\boldsymbol{\nu}_r)+\boldsymbol{C}_A(\boldsymbol{\nu}_r).$$

$$\boldsymbol{P}_{345}=\mathrm{diag}\{0,0,1,1,1,0\}$$

Since the equilibrium satisfies z = ϕ = θ = 0, only the heave–roll–pitch restoring terms remain, so the 3:5 block of $\mathbf{C}$ equals $\boldsymbol{G}$ and $\boldsymbol{C\xi}=\boldsymbol{G\eta}$ when $\boldsymbol{G}=\mathbf{P}_{345}\,\mathbf{C}\,\mathbf{P}_{345}$.

$$\dot{\boldsymbol{\eta}}=\boldsymbol{J}_\Theta(\boldsymbol{\eta})(\boldsymbol{\nu}_r+\boldsymbol{\nu}_c)$$
$$\boldsymbol{M}\dot{\boldsymbol{\nu}}_r+\boldsymbol{C}(\boldsymbol{\nu}_r)\boldsymbol{\nu}_r+\boldsymbol{D}\boldsymbol{\nu}_r+\boldsymbol{G}\boldsymbol{\eta}=\boldsymbol{\tau}+\boldsymbol{\tau}_{\mathrm{wave}}+\boldsymbol{\tau}_{\mathrm{wind}}$$

where $\boldsymbol{\nu}_r=\boldsymbol{\nu}-\boldsymbol{\nu}_c$ is the relative velocity vector.

$\boldsymbol{M_A}=\boldsymbol{A}_0^{\boldsymbol{eq}}$ and $\boldsymbol{D_A}=\boldsymbol{B}_0^{\mathrm{eq}}+\boldsymbol{B}_{\mathrm{v}}$ are evaluated at zero speed, while the velocity-dependent Coriolis and centripetal effects in the unified model are represented explicitly by $\boldsymbol{C_{RB}}(\boldsymbol{\nu_r})$and $\boldsymbol{C_A}(\boldsymbol{\nu_r})$.The speed-dependent matrices $\boldsymbol{A_U^{\mathrm{eq}}}$and $\boldsymbol{B_U^{\mathrm{eq}}}$are not used directly, since they already contain the linearized forward-speed couplings associated with the equilibrium speed *U*. Using them together with $\boldsymbol{C_{RB}}(\boldsymbol{\nu_r})$and $\boldsymbol{C_A}(\boldsymbol{\nu_r})$would therefore double count the Coriolis and centripetal effects.

Case Study: WAMIT Tanker Model

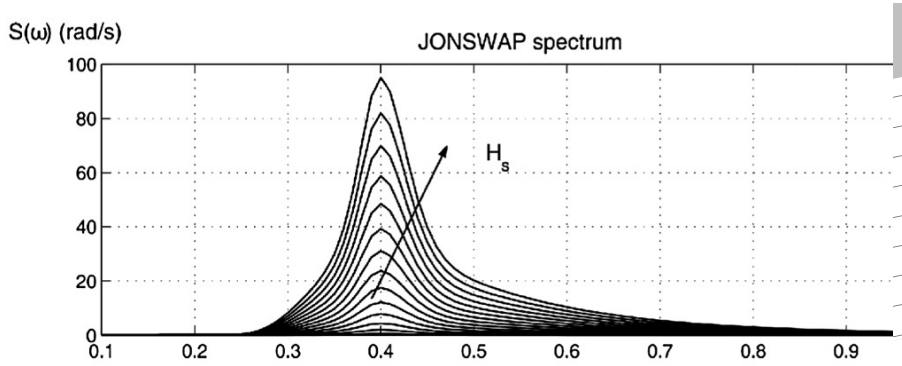
Fossen, T. I. (2025). Maneuvering Coefficient Estimation from Frequency-Dependent Added Mass and Damping: A Power-Based Approach. Oceanic Engineering 341, 122494. https://doi.org/10.1016/j.oceaneng.2025.122494

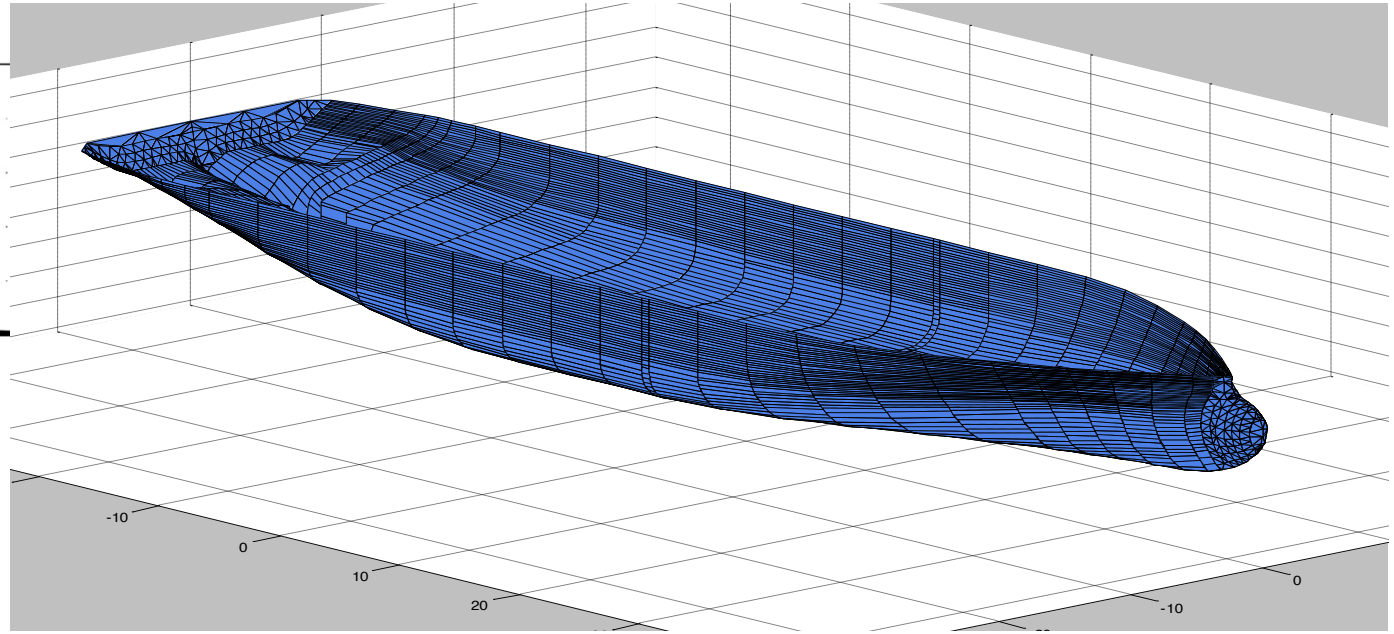
Case Study: WAMIT Tanker Model

The frequency-domain data of the tanker hull is obtained from the MSS toolbox, which provides added mass and damping coefficients, as well as first-order wave force RAOs generated by WAMIT.

The model represents a large crude oil tanker with a length between perpendiculars of L_{pp} = 246 m, B = 46 m, and T = 10 m. The displacement mass is m = 94,620 tonnes. The transverse and longitudinal metacentric heights are GM_T = 9.98 m and GM_L = 411.6 m.

1st-order wave forces were computed using frequency-dependent RAOs and a JONSWAP wave spectrum with H_s = 4 m, ω_p = 0.8 rad/s, and γ = 3.3.

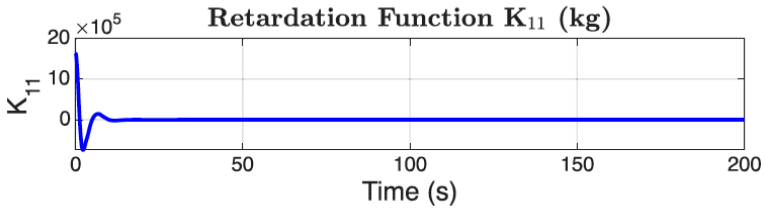


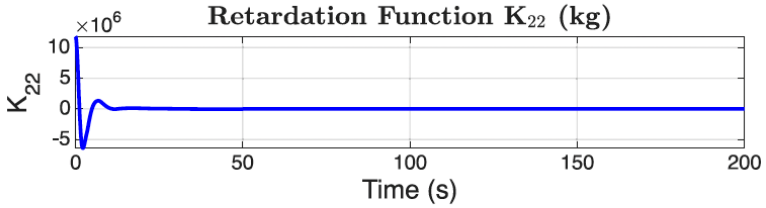


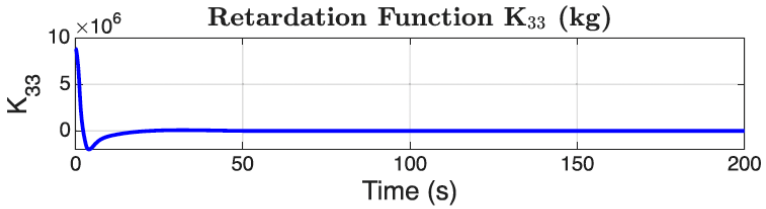
exManeuveringModel.m

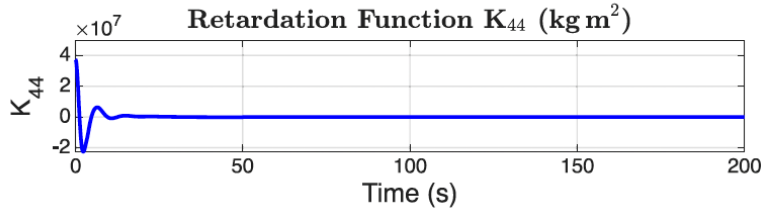
```
% This script simulates the response of a vessel in waves using Cummins (1962)
% equation and an equivalent maneuvering model (Fossen, 2025). It calculates
% the wave-induced forces using 1st-order force RAOs, solves the full hydrodynamic
% model including memory effects, and compares it with a simplified approximation
% using constant equivalent added mass and damping matrices (A_eq, B_eq) according
% according to:
%
%   A_eq = ∫ A_U(ω) S_N(ω) dω
%   B_eq = ∫ B_U(ω) S_N(ω) dω
%
% where
%
%   S(ω)                 : Wave energy spectrum
%   S_N(ω) = S(ω) / m_0  : Normalized wave spectrum
%   m_0 = ∫ S(ω) dω      : Zero spectral moment
%   A_U(ω) and B_U(ω)    : Added mass and damping matrices at speed U
%
% The integrals are evaluated numerically using trapezoidal integration.
% This ensures that the kinetic energy and power dissipation properties
% of the frequency-dependent system are preserved in the equivalent
% constant matrices.
%
% References:
%   Cummins, W. E. (1962). The impulse response function and ship motions.
%   Schiffstechnik, 9(47), 101–109.
%
%   Fossen, T. I. (2025). Maneuvering Coefficient Estimation from Frequency-
%   Dependent Added Mass and Damping: A Power-Based Approach.
%   Ocean Engineering, 341, 122494.
%
```

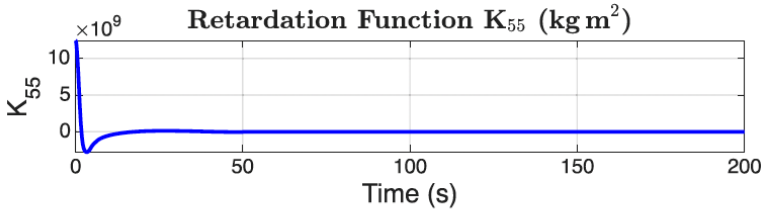
```
>> exManeuveringModel
Loaded the Tanker at 0.00 m/s
Spectrum: JONSWAP (no spreading function)
Significant wave height: Hs = 4 m
Zero crossing period: Tz = 5.5763 s
Peak period: T0 = Tz / 0.710 = 7.854 s
Peak frequency: w0 = 2*pi/T0 = 0.8 rad/s
Number of wave frequencies: 60
Number of wave directions: 1
>>
>> display(vessel.A_eq)
   1.0e+11 *
    0.0000         0   -0.0000         0   -0.0047         0
         0    0.0003         0    0.0010         0    0.0012
   -0.0000         0    0.0015         0   -0.0006         0
         0    0.0010         0    0.0954         0   -0.0458
   -0.0047         0   -0.0006         0    5.2530         0
         0    0.0012         0   -0.0460         0    1.2585
>>
>> display(vessel.B_eq)
   1.0e+11 *
    0.0000         0   -0.0000         0   -0.0045         0
         0    0.0003         0    0.0015         0    0.0029
   -0.0000         0    0.0004         0    0.0063         0
         0    0.0015         0    0.0200         0   -0.0161
   -0.0045         0    0.0063         0    1.8231         0
         0    0.0029         0   -0.0161         0    1.1094
>>
```

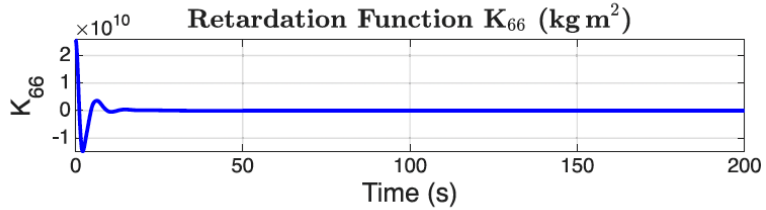

Retardation Function K_{11} (kg)
$\times 10^5$
K_{11}
Time (s)

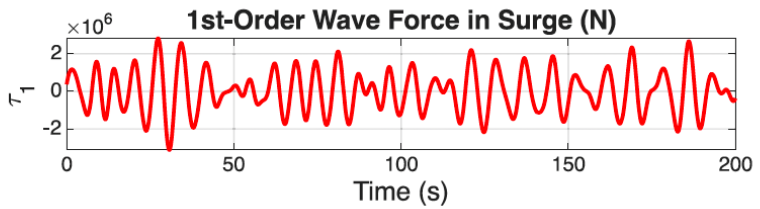

Retardation Function K_{22} (kg)
$\times 10^6$
K_{22}
Time (s)

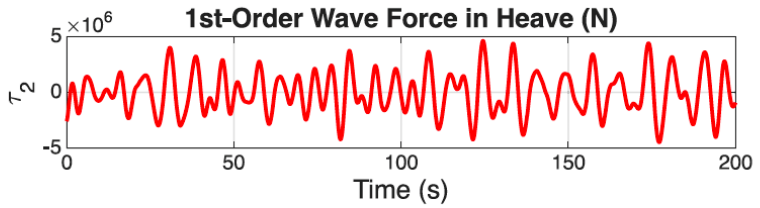

Retardation Function K_{33} (kg)
$\times 10^6$
K_{33}
Time (s)

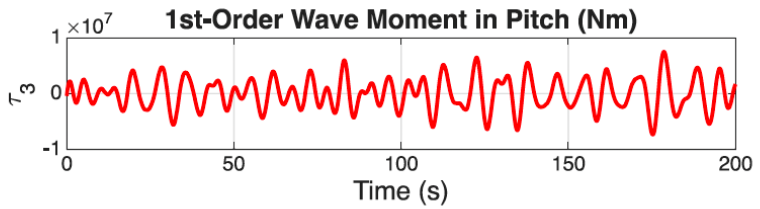

Retardation Function K_{44} (kg m^2)
$\times 10^7$
K_{44}
Time (s)

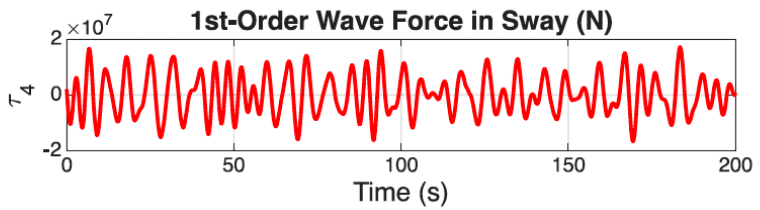

Retardation Function K_{55} (kg m^2)
$\times 10^9$
K_{55}
Time (s)

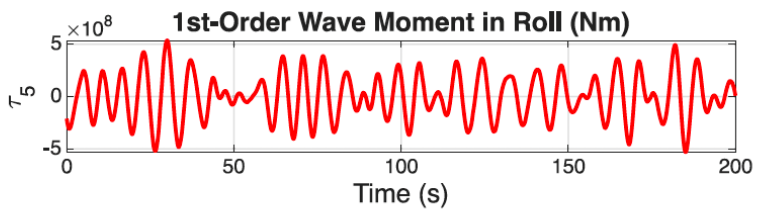

Retardation Function K_{66} (kg m^2)
$\times 10^{10}$
K_{66}
Time (s)

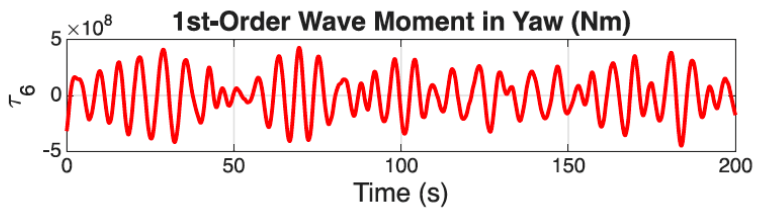

1st-Order Wave Force in Surge (N)
$\times 10^6$
τ_1
Time (s)

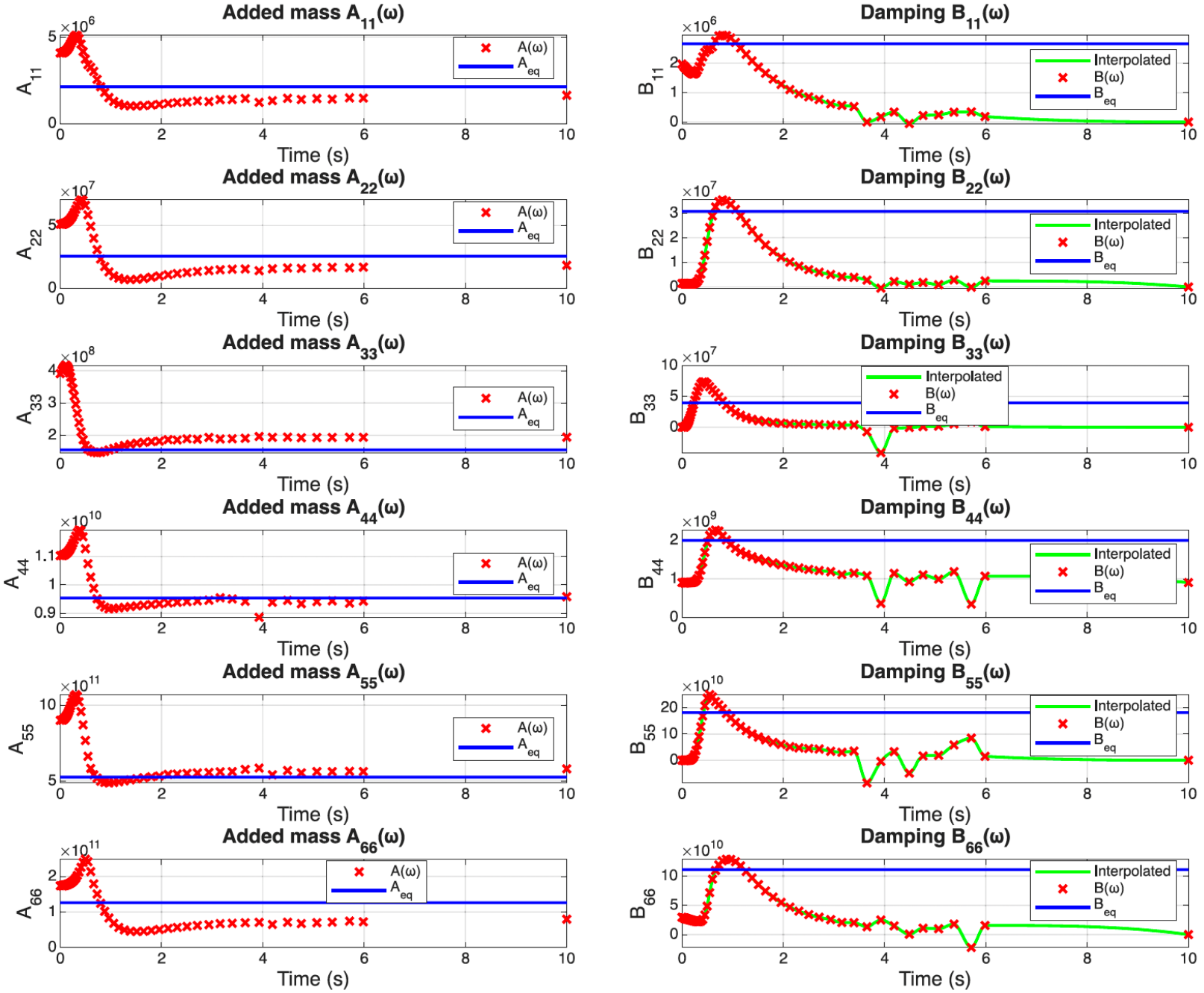

1st-Order Wave Force in Heave (N)
$\times 10^6$
τ_2
Time (s)

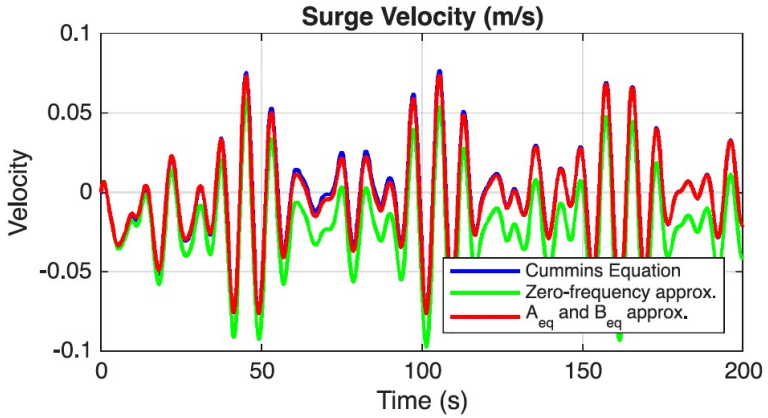

1st-Order Wave Moment in Pitch (Nm)
$\times 10^7$
τ_3
Time (s)

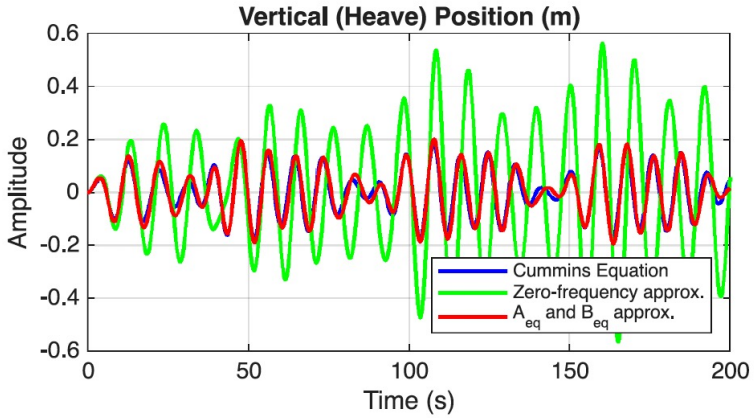

1st-Order Wave Force in Sway (N)
$\times 10^7$
τ_4
Time (s)

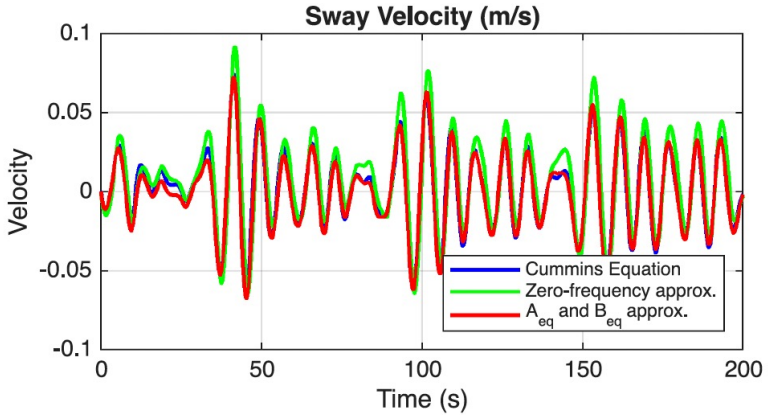

1st-Order Wave Moment in Roll (Nm)
$\times 10^8$
τ_5
Time (s)

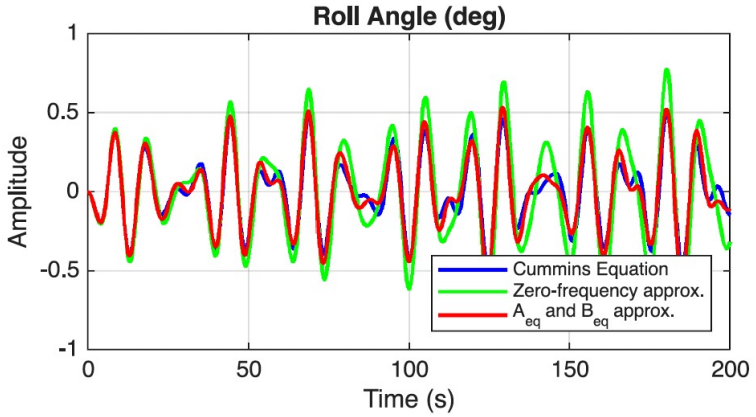

1st-Order Wave Moment in Yaw (Nm)
$\times 10^8$
τ_6
Time (s)

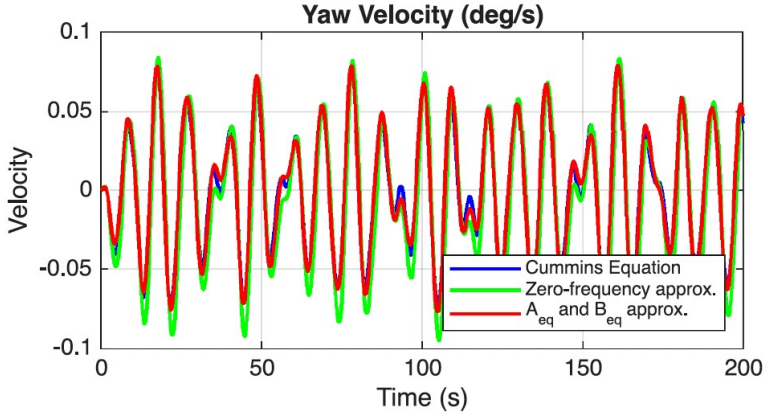

Added mass $A_{11}(\omega)$
Damping $B_{11}(\omega)$
Added mass $A_{22}(\omega)$
Damping $B_{22}(\omega)$
Added mass $A_{33}(\omega)$
Damping $B_{33}(\omega)$
Added mass $A_{44}(\omega)$
Damping $B_{44}(\omega)$
Added mass $A_{55}(\omega)$
Damping $B_{55}(\omega)$
Added mass $A_{66}(\omega)$
Damping $B_{66}(\omega)$
A(ω)
A_{eq}
Interpolated
B(ω)
B_{eq}
Time (s)

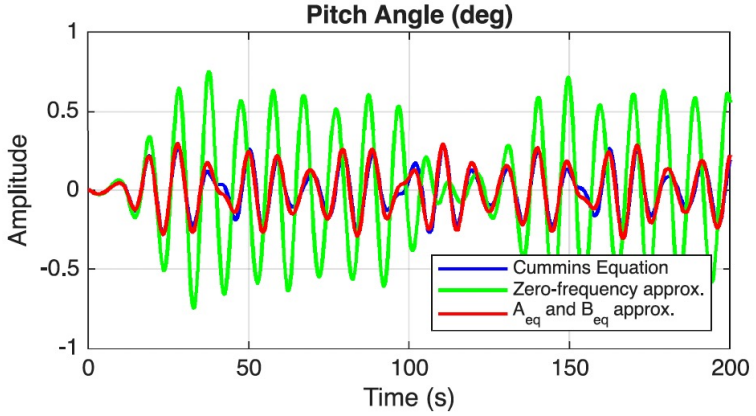

Surge Velocity (m/s)
Velocity
Time (s)
Cummins Equation
Zero-frequency approx.
A_{eq} and B_{eq} approx.


Vertical (Heave) Position (m)
Amplitude
Time (s)
Cummins Equation
Zero-frequency approx.
A_{eq} and B_{eq} approx.


Sway Velocity (m/s)
Velocity
Time (s)
Cummins Equation
Zero-frequency approx.
A_{eq} and B_{eq} approx.


Roll Angle (deg)
Amplitude
Time (s)
Cummins Equation
Zero-frequency approx.
A_{eq} and B_{eq} approx.


Yaw Velocity (deg/s)
Velocity
Time (s)
Cummins Equation
Zero-frequency approx.
A_{eq} and B_{eq} approx.


Pitch Angle (deg)
Amplitude
Time (s)
Cummins Equation
Zero-frequency approx.
A_{eq} and B_{eq} approx.

Summary: Case Study

Table 1

Percentage improvement (%) of the power-based averaging method over the conventional zero-frequency approximation, based on RMSE values computed over 10 min of simulation, using the Cummins formulation as reference. The S175 container ship is simulated at constant forward speed $U = 5.1$ m/s, while the tanker is at rest. The improvement is defined as $100 \times (\mathrm{RMSE}_{\mathrm{zero}} - \mathrm{RMSE}_{\mathrm{eq}})/\mathrm{RMSE}_{\mathrm{zero}}$, where RMSE is evaluated for each DOF using either position or velocity signals, depending on the motion type. The RMSE is calculated over the full 6-DOF time-domain motion signals (positions or velocities) for each DOF.

Test case		Six DOF					
Vessel	Condition (ω and H_s)	Surge	Sway	Heave	Roll	Pitch	Yaw
Tanker (WAMIT)	$\omega_p = 0.6$ rad/s, $H_s = 4$ m	94.5	88.1	96.0	85.5	90.6	92.0
	$\omega_p = 0.8$ rad/s, $H_s = 4$ m	97.1	79.1	91.9	69.8	91.0	72.6
	$\omega_p = 1.0$ rad/s, $H_s = 3$ m	97.5	78.5	70.3	68.0	85.3	89.7
S175 Container (ShipX)	$\omega_e = 0.73$ rad/s, $H_s = 4$ m	64.8	77.0	95.3	57.0	96.3	76.3
	$\omega_e = 1.04$ rad/s, $H_s = 4$ m	62.3	63.0	87.6	56.8	86.3	72.4
	$\omega_e = 1.37$ rad/s, $H_s = 3$ m	53.8	62.5	66.2	50.8	63.2	62.1

Fossen, T. I. (2025). Maneuvering Coefficient Estimation from Frequency-Dependent Added Mass and Damping: A Power-Based Approach. Oceanic Engineering 341, 122494. https://doi.org/10.1016/j.oceaneng.2025.122494

Summary: Zero-Frequency vs. Power-Based Method

The Well-Celebrated Zero-Frequency Assumption

- In maneuvering models (surge, sway and yaw), hydrodynamic added mass and damping are often taken from the zero-frequency limit of seakeeping coefficients:

$$\mathbf{A}_{ii}(0) \text{ and } \mathbf{B}_{ii}(0)$$

This assumption is only valid in surge, sway, and yaw (DOF 1, 2, 6), since the natural frequencies in heave, roll, and pitch (DOF 3, 4, 5) lie within the wave-frequency range.

- Justified since radiation damping vanishes at high frequency in surge, sway, yaw.
- Works reasonably well for 3-DOF maneuvering models where we neglect heave, roll and pitch.

The Problem for 6-DOF Coupled Motions

- In heave, roll, pitch (DOF 3, 4, 5), wave radiation is strongly frequency-dependent, and it can be close to the natural frequencies in the same modes.
- Using zero-frequency values:
  → Ignores coupling between horizontal motions (DOF 1, 2, 6) and vertical motions (DOF 3, 4, 5).
  → Leads to incorrect energy balance and unrealistic damping in coupled 6-DOF simulations.

The Power-Based Method

- Computes equivalent $\mathbf{A}^{eq}$ and $\mathbf{B}^{eq}$ by matching radiated wave power over the full frequency spectrum.
- Preserves the correct energy dissipation for all DOFs.
  → Ensures consistent coupling between horizontal and vertical motions.
- Provides a time-domain valid representation of hydrodynamics derived from frequency-domain data.

Chapter Goals - Revisited

- Distinguish frequency-domain seakeeping models from time-domain maneuvering models and explain when each is appropriate.
- Define the seakeeping perturbation coordinates in {s} and transform motions and forces between {s} and the BODY-fixed frame {b}.
- Interpret frequency-dependent added-mass and potential-damping matrices, hydrostatic restoring forces, and viscous-damping contributions.
- Explain how frequency-dependent added mass and damping are identified from forced-oscillation experiments using frequency-by-frequency least-squares fitting or computed using potential-flow programs.
- Explain why inserting frequency-dependent hydrodynamic coefficients directly into a time-domain differential equation causes an inconsistent mixing of time- and frequency-domain quantities.
- Formulate Cummins' equation using asymptotic added mass and retardation functions to represent fluid-memory effects.
- Explain when zero-frequency coefficients are reasonable for surge, sway, and yaw, and why this approximation can fail for coupled 6-DOF motion.
- Compute sea-state-dependent equivalent added-mass and damping matrices by spectral power-based averaging and use them in a BODY-fixed time-domain model.